

Preface

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Contents

Contents	i
1 Normed Linear Spaces	1
1.1 Basic Definitions and Results	1
1.2 Completeness and Compactness	3
Exercises	4
1.3 Linear Operators	5
1.4 Bounded Linear Functionals	7
Exercises	10
1.5 ℓ^p Spaces	13
Exercises	14
1.6 Consequences of The Baire Category Theorem	15
Exercises	21
1.7 Continuous Dual, Adjoint Maps, and Weak Convergence	22
Exercises	22
1.8 L^p Spaces	23
Exercises	23
2 Hilbert Spaces	24
2.1 Inner Product Spaces	24
Exercises	26
2.2 Hilbert Spaces	27
Exercises	33
2.3 Compact Operators	40

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Chapter 1

Normed Linear Spaces

§ 1.1. Basic Definitions and Results

Definition 1.1.1. A seminorm on an $\mathbb{F}$ -vector space X is a map:

$$\|\cdot\| : X \rightarrow [0, \infty)$$

satisfying:

- (1) $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{F}$ and $x \in X$ (*homogeneity*);
- (2) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$ (*triangle inequality*).

If the following condition is also satisfied:

- (3) $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$ (*positive definiteness*),

then we say $\|\cdot\|$ is a norm. The ordered pair $(X, \|\cdot\|)$ is called a normed linear space.

Remark. In most cases, one can take $\mathbb{F}$ to be $\mathbb{R}$ or $\mathbb{C}$ without any loss of generality.

Remark. A normed linear space induces a metric by the equation $d(x, y) := \|y - x\|$.

Remark. A normed linear space is finite dimensional if it is as a vector space; otherwise it is infinite dimensional.

Definition 1.1.2. Let $(X, \|\cdot\|)$ be a normed linear space.

- (1) The open ball centered around $x_0 \in X$ with radius $r > 0$ is denoted $B(x_0, r) := \{x \in X \mid \|x - x_0\| < r\}$.
- (2) The closed ball centered around $x_0 \in X$ with radius $r > 0$ is denoted $\overline{B}(x_0, r) := \{x \in X \mid \|x - x_0\| \leq r\}$.
- (3) The sphere centered around $x_0 \in X$ with radius $r > 0$ is denoted $S(x_0, r) := \{x \in X \mid \|x - x_0\| = r\}$.

Definition 1.1.3. Two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on a normed linear space X are said to be equivalent if there exists a constant $C \geq 1$ such that, for all $x \in X$:

$$\frac{1}{C} \|x\|_\beta \leq \|x\|_\alpha \leq C \|x\|_\beta.$$

Example 1.1.1. Let $x \in \mathbb{R}^n$, where $x := (x_1, \dots, x_n)$. Then:

$$\begin{aligned}\|x\|_2 &:= \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}}, \\ \|x\|_1 &:= \sum_{i=1}^n |x_i|, \\ \|x\|_\infty &:= \max_{i=1}^n |x_i|, \\ \|x\|_p &:= \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \quad p > 1\end{aligned}$$

are norms on $\mathbb{R}^n$.

Example 1.1.2. Let $a, b \in \mathbb{R}$ with $a < b$. The set of continuous real-valued functions on $[a, b]$ is denoted $C([a, b]) := \{f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$. Then:

$$\begin{aligned}\|f\|_\infty &:= \sup\{|f(x)| \mid x \in [a, b]\}, \\ \|f\|_2 &:= \left(\int_a^b |f(x)|^2 dx \right)^{\frac{1}{2}}, \\ \|f\|_p &:= \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}},\end{aligned}$$

are all norms on $C([a, b])$.

Definition 1.1.4. Let $(X, \|\cdot\|)$ be a normed linear space. A subset $V \subset X$ is convex if for all $x, y \in V$ and for all $t \in [0, 1]$, $tx + (1-t)y \in V$.

Example 1.1.3. Any subspace of a normed linear space is convex. Moreover, by the triangle inequality any open ball is convex.

Remark. Note that the *supremum norm* $\|\cdot\|_\infty$ on $C([a, b])$ only exists (by the Extreme Value Theorem) because $[a, b]$ is compact. Hence the above example can be generalized to a compact metric space Y . In particular, for $C(Y) := \{f : Y \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, the norm $\|f\|_\infty := \sup\{|f(x)| \mid x \in Y\}$ always exists and is finite.

Proposition 1.1.1. Let (X, d) and (Y, ρ) be metric spaces. Let $f : X \rightarrow Y$. Then f is continuous if and only if the preimage of an open set in Y is open in X .

Proof. Suppose f is continuous. Let $V \subseteq Y$ be open. If $f^{-1}(V) = \emptyset$, then there is nothing to show. If not, pick any $x \in f^{-1}(V)$. Then $f(x) \in V$. Since V is open, there exists $\epsilon > 0$ such that $B(f(x), \epsilon) \subseteq V$. Since f is continuous, there exists a $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon) \subseteq V$. Thus $B(x, \delta) \subseteq f^{-1}(V)$; i.e., $f^{-1}(V)$ is open.

Conversely, let $x \in X$ and $\epsilon > 0$. Since $B(f(x), \epsilon)$ is open in Y , $f^{-1}(B(f(x), \epsilon))$ is open in X . Clearly $x \in f^{-1}(B(f(x), \epsilon))$. Since this set is open, there exists $\delta > 0$ such that $x \in B(x, \delta) \subseteq f^{-1}(B(f(x), \epsilon))$. Hence $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$. Since the set of all open balls is a basis for a topology, any open set can be written as the union of open balls. Thus f is continuous. $\square$

Remark. It follows that f is continuous if and only if the preimage of a closed set is closed.

Definition 1.1.5. Let $(X, \|\cdot\|)$ be a normed linear space and $\{x_n\}_{n=1}^{\infty}$ be a sequence in X .

- (1) The sequence $\{x_n\}_{n=1}^{\infty}$ converges to $x \in X$ in the $\|\cdot\|$ -norm if $\|x - x_n\| \rightarrow 0$.
- (2) A series in X is a formal sum $\sum_{n=1}^{\infty} x_n$. We say the series $\sum_{n=1}^{\infty} x_n$ converges to $x \in X$ if the sequence of partial sums $\{\sum_{i=1}^n x_i\}_{n=1}^{\infty}$ converges to x . We say $\sum_{n=1}^{\infty} x_n$ is absolutely convergent if $\sum_{n=1}^{\infty} \|x_n\|$ converges.

Proposition 1.1.2. Let X be a normed linear space. The following are equivalent:

- (1) X is a Banach space;
- (2) Every absolutely convergent series in X converges.

Proof.

□

§ 1.2. Completeness and Compactness

Lemma 1.2.1. In $(\mathbb{R}^n, \|\cdot\|_{\infty})$, every closed ball $\overline{B}(0, c)$ is compact.

Proof.

□

Theorem 1.2.1. Let X be a finite dimensional normed linear space. Every closed and bounded subset of X is compact.

Proof.

□

Corollary 1.2.1. Every finite dimensional normed linear space is Banach.

Proof. Let X be a finite dimensional normed linear space and $\{x_n\}_{n=1}^{\infty}$ a Cauchy sequence. Then $\{x_n\}_{n=1}^{\infty}$ is closed and bounded, hence it is compact. Thus $\{x_n\}_{n=1}^{\infty}$ admits a convergent subsequence. But since $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence which admits a convergent subsequence, $\{x_n\}_{n=1}^{\infty}$ is convergent. Thus X is Banach. □

Corollary 1.2.2. Every finite dimensional subspace of a normed linear space is closed.

Proof. This follows from Corollary 1.2.1 and Exercise 1.2.4. □

Lemma 1.2.2 (Riesz's Lemma). *Let $(X, \|\cdot\|)$ be a normed linear space. Let Z be a closed and proper subspace of X . Let $0 < \lambda < 1$ be fixed. Then there exists some $x \in X \setminus Z$ such that $\|x\| = 1$ and $\text{dist}(x, Z) > \lambda$.*

Proof.

□

Theorem 1.2.2. *Let X be a normed linear space. Then X is finite dimensional if and only if $\overline{B}(0, 1)$ is compact.*

Proof.

□

Exercises

Exercise 1.2.1. *Let K be a compact metric space (fine to assume that $K = [0, 1]$). Show that $C(K)$ paired with the sup-norm $\|\cdot\|_\infty$ is Banach.*

Proof.

□

Exercise 1.2.2. *In any metric space, a closed subset of a compact set is compact.*

Proof.

□

Exercise 1.2.3. *In any metric space, a Cauchy sequence which has a convergent subsequence must converge.*

Proof.

□

Exercise 1.2.4. *A subspace of a Banach space is closed if and only if it is Banach.*

Proof. Let X be a Banach space and Y a subspace. Suppose that Y is closed. Let $\{y_n\}_{n=1}^\infty \subseteq Y$ be a Cauchy sequence. Then $\{y_n\}_{n=1}^\infty$ is Cauchy in X . Therefore $y_n \rightarrow x$ for some $x \in X$. But since Y is closed, it contains all of its limit points. Hence $x \in Y$. Therefore $\{y_n\}_{n=1}^\infty$ converges in Y , establishing that Y is Banach.

Conversely, suppose that Y is Banach. Let $\{y_n\}_{n=1}^\infty \subseteq Y$ be a sequence such that $y_n \rightarrow x \in X$. Note that $\{y_n\}_{n=1}^\infty$ is convergent, hence Cauchy. But since Y is Banach, $\{y_n\}_{n=1}^\infty$ converges to some $y \in Y$. It must be the case that $x = y \in Y$. Therefore Y contains all of its limit points; i.e., it is closed in X . □

Exercise 1.2.5. In any normed linear space X with closed subspace Z , if $x \in X \setminus Z$ then $\text{dist}(x, Z) > 0$.

Proof. Suppose towards contradiction $\text{dist}(x, Z) = \inf\{\|x - z\| \mid z \in Z\} = 0$. We can find a sequence $\{z_n\}_{n=1}^\infty \subseteq Z$ such that $\|x - z_n\| \rightarrow 0$. Hence $z_n \rightarrow x$. But since Z is closed, $x \in Z$. This is a contradiction, hence $\text{dist}(x, Z) > 0$. $\square$

Exercise 1.2.6. In any normed linear space X with subspace Z , if $x \in X$ and $z \in Z$ then $\text{dist}(x, Z) = \text{dist}(x + z, Z)$.

Proof. Note that $\text{dist}(x + z, Z) = \inf\{\|(x + z) - z_0\| \mid z_0 \in Z\} = \inf\{\|x - (z_0 - z)\| \mid z_0 \in Z\}$. Since $\{z_0 - z \mid z_0 \in Z\} = Z$, we have that $\inf\{\|x - z\| \mid z \in Z\} = \inf\{\|(x + z) - z_0\| \mid z_0 \in Z\}$; i.e., $\text{dist}(x, Z) = \text{dist}(x + z, Z)$. $\square$

Exercise 1.2.7. Let X and Y be normed linear spaces and let $\varphi : X \rightarrow Y$ be a linear operator. Show that φ is continuous if and only if it is continuous at 0.

Proof. $\square$

§ 1.3. Linear Operators

Definition 1.3.1. Let X and Y be normed linear spaces. A map $T : X \rightarrow Y$ is linear if $T(x + \alpha y) = T(x) + \alpha T(y)$ for all $x, y \in X$ and $\alpha \in \mathbb{F}$. T is called a linear transformation (or linear operator). If X is a normed linear space over $\mathbb{F}$, then a linear transformation $\varphi : X \rightarrow \mathbb{F}$ is called a linear functional.

Remark. The norm on $\mathbb{F}$ is the usual absolute value in $\mathbb{R}$ or modulus operator in $\mathbb{C}$.

Example 1.3.1. Linear operators between $\mathbb{R}^n$ and $\mathbb{R}^m$ correspond to $n \times m$ matrices.

Example 1.3.2. Consider the normed linear space $(C([0, 1]), \|\cdot\|_\infty)$. Evaluation at zero is a linear functional, as well as the map $x \mapsto \int_0^1 x(t) dt$. Moreover, $T : C([0, 1]) \rightarrow C([0, 1])$ defined by $x(t) \mapsto \int_0^1 K(s, t)x(s) ds$ is a linear transformation, where K is a continuous function on $[0, 1] \times [0, 1]$.

Definition 1.3.2. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ a linear transformation. T is bounded if there exists $M > 0$ such that for all $x \in X$, $\|x\| \leq 1$ implies $\|Tx\| \leq M$.

Theorem 1.3.1. Let X and Y be normed linear spaces. A linear transformation $T : X \rightarrow Y$ is bounded if and only if it is continuous.

Proof. Suppose T is continuous. Then T is continuous at zero. Choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$. Now let $x \in X$ with $\|x\| \leq 1$. Then $\|Tx\| = \left\| T\left(\frac{x\delta}{\delta}\right) \right\| = \frac{1}{\delta} \|T(\delta x)\| \leq \frac{1}{\delta}$. Therefore T is bounded.

Conversely, let $M > 0$ satisfy $\|Tx\| \leq M$ for all $x \in \overline{B}(0, 1)$. Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{M}$. Then $\|x\| \leq \delta$ implies $\|\frac{M}{\epsilon}x\| \leq 1$. Hence by our assumption $\|T(\frac{M}{\epsilon}x)\| \leq M$, which is equivalent to $\|Tx\| \leq \epsilon$. Thus T is continuous at 0, hence it is continuous everywhere. $\square$

Example 1.3.3. Consider $D : C^1([0, 1]) \rightarrow C([0, 1])$ given by $Df = f'$. Consider the sequence of functions $\{f_n\}_{n=1}^\infty \subseteq C^1([0, 1])$ given by $f_n(x) = \frac{1}{\sqrt{n}}x^n$. Then for all $n \in \mathbb{N}$, $Df_n(x) = \sqrt{n}x^{n-1}$. Hence D is unbounded.

Theorem 1.3.2. Let X be a normed linear space. A linear functional $\varphi : X \rightarrow \mathbb{F}$ is continuous if and only if its kernel is closed.

Proof. Suppose φ is continuous. Then $\ker \varphi = \varphi^{-1}(\{0\})$ is closed, as the continuous preimage of a closed set is closed.

Conversely, suppose towards contradiction φ is not continuous. Then φ is unbounded; i.e., there exists a sequence $\{x_n\}_{n=1}^\infty \subset X$ such that $\|x_n\| = 1$ for all $n \in \mathbb{N}$ and $|\varphi(x_n)| \rightarrow \infty$. If $\ker \varphi = X$ then there is nothing to show. Assume $\ker \varphi \neq X$ and pick some $x \in X \setminus \ker \varphi$. Define $y_n := x - \frac{\varphi(x)}{\varphi(x_n)}x_n$. Then $\varphi(y_n) = \varphi(x) - \frac{\varphi(x)}{\varphi(x_n)}\varphi(x_n) = 0$. Hence $y_n \in \ker \varphi$ for all $n \in \mathbb{N}$. But note that $\|y_n - x\| = \left\| -\frac{\varphi(x)}{\varphi(x_n)}x_n \right\| = \frac{|\varphi(x)|}{|\varphi(x_n)|}\|x_n\| \rightarrow 0$. Hence $y_n \rightarrow x$. Since $\ker \varphi$ is closed, $x \in \ker \varphi$. This is a contradiction, establishing the proof. $\square$

Theorem 1.3.3. Let X be a finite dimensional normed linear space and Y a normed linear space. If $T : X \rightarrow Y$ is linear, then T is continuous.

Proof. Pick a basis for X such that $X = \text{span}\{b_1, \dots, b_n\}$. Define $M := \max_{i=1}^n \{T(b_i)\}$. For each $j = 1, \dots, n$, define a linear functional $\varphi_j : X \rightarrow \mathbb{F}$ by:

$$\begin{aligned} \varphi_j(x) &= \varphi_j\left(\sum_{i=1}^n \alpha_i b_i\right) \\ &= \sum_{i=1}^n \alpha_i \varphi_j(b_i) \\ &:= \alpha_j \end{aligned}$$

Since all subspaces of X are closed by Corollary 1.2.2, surely $\ker \varphi_j$ is closed. Hence by Theorem 1.3.2 φ_j is continuous. Since φ_j is continuous, find $L > 0$ sufficiently large enough

such that for every $x \in X$ with $\|x\| \leq 1$, we have $|\varphi_i(x)| \leq L$. Since $T : X \rightarrow Y$ is linear:

$$\begin{aligned} \|Tx\| &= \left\| T \sum_{i=1}^n \alpha_i b_i \right\| \\ &= \left\| \sum_{i=1}^n \alpha_i T b_i \right\| \\ &= \left\| \sum_{i=1}^n \varphi_i(x) T b_i \right\| \\ &\leq \sum_{i=1}^n |\varphi_i(x)| \|T b_i\| \\ &\leq nLM \\ &< \infty. \end{aligned}$$

Thus T is continuous. □

Theorem 1.3.4. *Let X be a finite dimensional vector space. All norms on X are equivalent.*

Proof. Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be two norms. Consider $\text{id} : (X, \|\cdot\|_\alpha) \rightarrow (X, \|\cdot\|_\beta)$. This map is surely linear, hence by Theorem 1.3.3 id is continuous. Hence we can find some $A > 0$ sufficiently large such that, for all $x \in X$ with $\|x\|_\alpha \leq 1$, we have $\|x\|_\beta \leq A$. Thus:

$$\begin{aligned} \|x\|_\beta &= \left\| \|x\|_\alpha \frac{x}{\|x\|_\alpha} \right\|_\beta \\ &= \|x\|_\alpha \left\| \frac{x}{\|x\|_\alpha} \right\|_\beta \\ &\leq \|x\|_\alpha A. \end{aligned}$$

Applying the same argument to id^{-1} gives $\|x\|_\alpha \leq \|x\|_\beta B$, establishing the theorem. □

§ 1.4. Bounded Linear Functionals

Definition 1.4.1. Let X and Y be normed linear spaces.

- (1) The space of bounded linear operators between X and Y is denoted $\mathcal{L}(X, Y) := \{T : X \rightarrow Y \mid T \text{ is bounded}\}$.
- (2) The operator norm of a bounded linear operator $T \in \mathcal{L}(X, Y)$ is $\|T\|_{\mathcal{L}(X, Y)} := \sup\{\|Tx\|_Y \mid x \in \overline{B}_X(0, 1)\}$.
- (3) The continuous dual of X is defined as $X^* := \mathcal{L}(X, \mathbb{F})$.

Remark. We denote $\mathcal{L}(X, X)$ by $\mathcal{L}(X)$.

Proposition 1.4.1. *If $T \in \mathcal{L}(X, Y)$ and $x \in X$, then $\|Tx\| \leq \|T\|_{\mathcal{L}(X, Y)} \cdot \|x\|$.*

Proof. If $x = 0$, then there is nothing to show. If $x \neq 0$, then:

$$\begin{aligned}\|Tx\| &= \left\| \|x\| T\left(\frac{x}{\|x\|}\right) \right\| \\ &= \left\| T\left(\frac{x}{\|x\|}\right) \right\| \|x\| \\ &\leq \|T\|_{\mathcal{L}(X,Y)} \|x\|. \quad \square\end{aligned}$$

Proposition 1.4.2. *If $S, T \in \mathcal{L}(X)$, then $S \circ T \in \mathcal{L}(X)$.*

Proof. Let $x \in X$ such that $\|x\| \leq 1$. By Proposition 1.4.1:

$$\begin{aligned}\|(S \circ T)(x)\| &\leq \|S(T(x))\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|Tx\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)} \|x\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)}.\end{aligned}$$

Taking the supremum over all $\|x\| \leq 1$ gives $\|S \circ T\|_{\mathcal{L}(X)} \leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)}$. $\square$

Theorem 1.4.1. *Let X be a normed linear space and Y a Banach space. Then $(\mathcal{L}(X, Y), \|\cdot\|_{\mathcal{L}(X,Y)})$ is Banach.*

Proof. Let $\{T_n\}_{n=1}^\infty \subseteq \mathcal{L}(X, Y)$ be Cauchy. Note that for any $x \in X$, $\{T_n(x)\}_{n=1}^\infty$ will be Cauchy in Y . Since Y is complete, this sequence has a limit. Define $Tx := \lim_{n \rightarrow \infty} T_n x$. We must show that T is linear, bounded, and $T_n \rightarrow T$.

Let $x, y \in X$, $\alpha \in \mathbb{F}$, and $n \in \mathbb{N}$. Then $T_n(x + \alpha y) = T_n x + \alpha T_n y$. By limit laws, $T(x + \alpha y) = Tx + \alpha Ty$. Thus T is linear.

Since $\{T_n\}_{n=1}^\infty$ is Cauchy in $\mathcal{L}(X, Y)$, this means $\{T_n\}_{n=1}^\infty$ is bounded. Hence we can find $M > 0$ sufficiently large such that $\|T_n\| \leq M$ for all $n \in \mathbb{N}$. For every $x \in X$ with $\|x\| \leq 1$, we have:

$$\begin{aligned}\|Tx\| &= \left\| \lim_{n \rightarrow \infty} T_n x \right\| \\ &= \lim_{n \rightarrow \infty} \|T_n x\| \\ &\leq \lim_{n \rightarrow \infty} \|T_n\|_{\mathcal{L}(X,Y)} \|x\| \\ &\leq \lim_{n \rightarrow \infty} \|T_n\|_{\mathcal{L}(X,Y)} \\ &\leq M.\end{aligned}$$

Thus T is bounded.

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ large such that $n, m \geq N$ implies $\|T_n - T_m\|_{\mathcal{L}(X,Y)} < \epsilon$. Let $x \in X$ such that $\|x\| \leq 1$. Then:

$$\begin{aligned}\|T_n x - T_m x\| &\leq \|(T_n - T_m)x\| \\ &\leq \|T_n - T_m\|_{\mathcal{L}(X,Y)} \|x\| \\ &\leq \|T_n - T_m\|_{\mathcal{L}(X,Y)} \\ &< \epsilon.\end{aligned}$$

Take $m \rightarrow \infty$. Then $\|T_n x - Tx\| \leq \epsilon$. Hence $\|(T_n - T)x\| \leq \epsilon$. Taking the supremum over all $x \in \overline{B}_X(0, 1)$ gives $\|T_n - T\|_{\mathcal{L}(X, Y)} \leq \epsilon$. Thus $T_n \rightarrow T$ in the operator norm. $\square$

Corollary 1.4.1 (Neumann's Theorem). *Let X be a Banach space and $T \in \mathcal{L}(X)$. If $\|T\| < 1$, then $I - T$ is invertible, and $(I - T)^{-1} = \sum_{n=0}^{\infty} T^n$.*

Proof. $\square$

Question. For any normed linear space X , is X^* nonempty? How can one construct an arbitrary bounded linear functional on X ?

Lemma 1.4.1 (Zorn's Lemma). *In a partially ordered set, if every chain has a maximal element, then the partially ordered set has a maximal element.*

Theorem 1.4.2 (Hahn-Banach). *Let X be a $\mathbb{R}$ -vector space and $p : X \rightarrow \mathbb{R}$ a map satisfying:*

$$\begin{aligned} p(x + y) &\leq p(x) + p(y), \\ p(\lambda x) &= \lambda p(x), \text{ provided } \lambda > 0. \end{aligned}$$

Let $Y \subseteq X$ be a subspace and $\varphi : Y \rightarrow \mathbb{R}$ a linear functional with:

$$\varphi(y) \leq p(y) \text{ for all } y \in Y.$$

Then there exists a linear functional $\Phi : X \rightarrow \mathbb{R}$ such that $\Phi|_Y = \varphi$ and:

$$\Phi(x) \leq p(x) \text{ for all } x \in X.$$

Proof. $\square$

Corollary 1.4.2. *Let X be a normed linear space and $Y \subseteq X$ a subspace. If $\varphi \in Y^*$, then there exists an extension $\Phi \in X^*$ such that $\|\varphi\|_{Y^*} = \|\Phi\|_{X^*}$.*

Proof. Define $p : X \rightarrow \mathbb{R}$ by $p(x) := \|\varphi\|_{Y^*} \|x\|$. Clearly $p(x + y) \leq p(x) + p(y)$ and $p(\lambda x) = \lambda p(x)$ provided $\lambda > 0$. Moreover:

$$\begin{aligned} p(y) &= \|\varphi\|_{Y^*} \|y\| \\ &\geq |\varphi(y)| \\ &\geq \varphi(y). \end{aligned}$$

By the Hahn-Banach Theorem, there exists an extension $\Phi : X \rightarrow \mathbb{R}$ such that $\Phi|_Y = \varphi$ and $\Phi(x) \leq p(x)$. From this, we have the following inclusion of sets:

$$\left\{ \left| \varphi\left(\frac{y}{\|y\|}\right) \right| \mid y \in Y, y \neq 0 \right\} \subseteq \left\{ \left| \Phi\left(\frac{x}{\|x\|}\right) \right| \mid x \in X, x \neq 0 \right\}.$$

Hence $\|\varphi\|_{Y^*} \leq \|\Phi\|_{X^*}$. Conversely, by our construction of p we have:

$$\begin{aligned} |\Phi(x)| &\leq |p(x)| \\ &\leq \|\varphi\|_{Y^*} \|x\|. \end{aligned}$$

Rearranging gives $\left| \Phi\left(\frac{x}{\|x\|}\right) \right| \leq \|\varphi\|_{Y^*}$. Taking the supremum over all $x \in X, x \neq 0$ gives $\|\Phi\|_{X^*} \leq \|\varphi\|_{Y^*}$. Having established equality, we've finally shown $\Phi \in X^*$. $\square$

Corollary 1.4.3. *Let X be a normed linear space and $Y \subseteq X$ a subspace. If $w \in X \setminus Y$ such that $\text{dist}(w, Y) > 0$, then there exists a bounded linear functional $\varphi \in X^*$ such that $\varphi(w) = 1$, $\varphi(y) = 0$ for all $y \in Y$, and $\|\varphi\|_{X^*} = \frac{1}{\text{dist}(w, Y)}$.*

Proof. Define $\varphi : \text{span}\{Y, \{w\}\} \rightarrow \mathbb{R}$ by $\varphi(y + \lambda w) = \lambda$. This is a linear functional, and:

$$\begin{aligned} \|\varphi\| &= \sup \left\{ \left| \varphi\left(\frac{y + \lambda w}{\|y + \lambda w\|}\right) \right| : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{|\lambda|}{\|y + \lambda w\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{1}{\left\| \frac{y}{\lambda} + w \right\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{1}{\|w - y\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\}^1 \\ &= \frac{1}{\inf\{\|w - y\| \mid y \in Y\}}^2 \\ &= \frac{1}{\text{dist}(w, Y)}. \end{aligned}$$

Applying Corollary 1.4.2 establishes the proof. $\square$

Remark. The previous corollary establishes that if $X = \{0\}$, then $X^* \neq \{0\}$.

Exercises

Exercise 1.4.1. *Let X be a normed linear space. Prove the limit laws: if $x_n \rightarrow x$, $y_n \rightarrow y$, and $\lambda \in \mathbb{F}$, then $x_n + \lambda y_n \rightarrow x + \lambda y$.*

Proof. $\square$

Exercise 1.4.2. *Let X be a normed linear space with norm $\|\cdot\|$. Prove that the map $\|\cdot\| : X \rightarrow \mathbb{R}$ is continuous.*

¹Since $\{\frac{y}{\lambda} \mid y \in Y, \lambda \in \mathbb{R}\} = Y$.

²Let $x_n \rightarrow \inf A$. Then $\frac{1}{x_n} \rightarrow \frac{1}{\inf A}$. Since $\frac{1}{\inf A}$ is an upper-bound and an adherent point of $\frac{1}{A}$, we have $\frac{1}{\inf A} = \sup \frac{1}{A}$.

Proof. Let $x, y \in X$. Then $\|x\| \leq \|x - y\| + \|y\|$, giving $\|x\| - \|y\| \leq \|x - y\|$. A similar tactic gives $\|y\| - \|x\| \leq \|x - y\|$. Hence $|\|x\| - \|y\|| \leq \|x - y\|$. This means $\|\cdot\|$ is Lipschitz, implying that it is continuous. $\square$

Exercise 1.4.3. Prove that the closure of a linear subspace in a normed linear space is also a subspace.

Proof. Let X be a normed linear space and Y a subspace. Let $u, v \in \bar{Y}$ and $\alpha \in \mathbb{F}$. There exists sequences $\{u_n\}_{n=1}^\infty, \{v_n\}_{n=1}^\infty$ such that $u_n \rightarrow u$ and $v_n \rightarrow v$. Hence $u_n + \alpha v_n \rightarrow u + \alpha v \in \bar{Y}$. Thus $\bar{Y}$ is a linear subspace. $\square$

Exercise 1.4.4. Prove that the operator norm has the three properties required of a norm

Proof. □

Exercise 1.4.5. Prove directly that if T is an unbounded linear operator, then it is discontinuous at 0.

Proof. Let $T : X \rightarrow Y$ be an unbounded linear operator. Since T is unbounded:

$$\|T\|_{\mathcal{L}(X,Y)} = \sup\{\|Tx\|_Y \mid x \in \bar{B}_X(0,1)\} = \infty.$$

We can find a sequence $\{x_n\}_{n=1}^\infty$ such that $\|x_n\|_X \leq 1$ and $\|Tx_n\|_Y \rightarrow \infty$. Consider the sequence $\left\{\frac{x_n}{\|Tx_n\|_Y}\right\}_{n=1}^\infty$. Clearly $\frac{x_n}{\|Tx_n\|_Y} \rightarrow 0$. But:

$$\left\|T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\| = \frac{\|Tx_n\|_Y}{\|Tx_n\|_Y} = 1.$$

Hence the sequence $\left\{T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\}_{n=1}^\infty$ does not converge to 0. Thus T is discontinuous at 0. $\square$

Exercise 1.4.6. Prove that the norm of a linear transformation is the infimum of all numbers M that satisfy the inequality $\|Tx\| \leq M\|x\|$ for all x .

Proof. Let M be such that $\|Tx\| \leq M\|x\|$ for all $x \in X$. Then $\frac{\|Tx\|}{\|x\|} \leq M$ for all $x \in X, x \neq 0$. Taking the supremum over all $x \in X, x \neq 0$ gives $\|T\|_{\mathcal{L}(X,Y)} \leq M$. Hence $\|T\|_{\mathcal{L}(X,Y)} \leq \inf\{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$.

Conversely, let $x \in X, x \neq 0$. Then $\|Tx\| \leq \|T\|_{\mathcal{L}(X,Y)}\|x\|$. Thus $\|T\|_{\mathcal{L}(X,Y)} \in \{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$. Surely $\|T\|_{\mathcal{L}(X,Y)} \geq \inf\{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$, establishing equality. $\square$

Exercise 1.4.7. On the space $C([0,1])$ we define "point-evaluation functionals" by $t^*(f) = f(t)$. Here $t \in [0,1]$ and $f \in C([0,1])$. Prove that $\|t^*\| = 1$. Prove that if $\varphi = \sum_{i=1}^n \lambda_i t_i^*$, where $t_1, t_2, \dots, t_n$ are distinct points in $[0,1]$, then $\|\varphi\| = \sum_{i=1}^n |\lambda_i|$.

Proof.

□

Exercise 1.4.8. *Prove that if a linear transformation maps some nonvoid open set of the domain space to a bounded set in the range space, then it is continuous.*

Proof. Let $T : X \rightarrow Y$ be linear. Let $U \subseteq X$ be open and nonvoid. Then there exists some $x \in U$. Find $r > 0$ such that $B(x, r) \subseteq U$.

Let $x_0 \in B(0, 1)$. Then $x + rx_0 \in B(x, r) \subseteq U$. Hence $T(x + rx_0) \in T(U)$. We can write:

$$\|Tx_0\| \leq \frac{\|T(x + rx_0)\| + \|Tx\|}{r}.$$

Since $T(U)$ is bounded, it must be the case that $\|Tx_0\| \leq M$ for some $M > 0$. Thus $T(B(0, 1))$ is bounded.

Now let $x \in X$. We have that:

$$\begin{aligned} \|Tx\| &= \left\| T\left(\frac{x}{2\|x\|} \cdot 2\|x\|\right) \right\| \\ &= 2 \left\| T\left(\frac{x}{2\|x\|}\right) \right\| \|x\| \\ &\leq 2M \|x\|. \end{aligned}$$

Thus T is continuous.

□

Exercise 1.4.9. *Let Y be a linear subspace in a normed linear space X . For a fixed $x \in X$, prove that:*

$$\text{dist}(x, Y) = \sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\}.$$

Proof. Let $\varphi \in X^*$ such that $\varphi|_Y = 0$ and $\|\varphi\| = 1$. Observe that:

$$\begin{aligned} \varphi(x) &= \varphi(x) - \varphi(y) \\ &= \varphi(x - y) \\ &\leq \|\varphi\| \|x - y\| \\ &= \|x - y\|. \end{aligned}$$

Taking the infimum over all $y \in Y$ gives $\varphi(x) \leq \text{dist}(x, Y)$. Taking the supremum over all $\varphi \in X^*$ such that $\varphi|_Y = 0$ and $\|\varphi\| = 1$ gives $\sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\} \leq \text{dist}(x, Y)$.

For the converse direction, we proceed by cases on $\text{dist}(x, Y)$. If $\text{dist}(x, Y) = 0$, then by the above inequality we have:

$$\begin{aligned} 0 &= \text{dist}(x, Y) \\ &\geq \sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\} \\ &\geq 0. \end{aligned}$$

Hence $\text{dist}(x, Y) = \sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\}$. Now suppose $\text{dist}(x, Y) > 0$. By Corollary 1.4.3 there exists a linear functional $\psi \in X^*$ such that $\psi(x) = 1$, $\psi|_Y = 0$, and

$\|\psi\| = \frac{1}{\text{dist}(x, Y)}$. Define a linear functional $\Phi : X \rightarrow \mathbb{F}$ by $\Phi(x_0) = \text{dist}(x_0, Y)\psi(x_0)$. Note that $\Phi|_Y = 0$ and:

$$\|\Phi\| = \text{dist}(x, Y)$$

$$\|\Phi\| = \text{dist}(x, Y)$$

□

§ 1.5. ℓ^p Spaces

Definition 1.5.1.

(1) Let $1 \leq p < \infty$. We define $\ell^p := \{x : \mathbb{N} \rightarrow \mathbb{F} \mid \sum_{j=1}^{\infty} |x_j|^p < \infty\}$. Likewise, define

$$\|\cdot\|_{\ell^p} : \ell^p \rightarrow \mathbb{F} \text{ by } \|x\|_{\ell^p} = \left(\sum_{j=1}^{\infty} |x_j|^p \right)^{\frac{1}{p}}.$$

(2) Let $p = \infty$. We define $\ell^\infty := \{x : \mathbb{N} \rightarrow \mathbb{F} \mid \sup_{n \in \mathbb{N}} |x_n| < \infty\}$. Likewise, define

$$\|\cdot\|_{\ell^\infty} : \ell^\infty \rightarrow \mathbb{F} \text{ by } \|x\|_{\ell^\infty} = \sup_{n \in \mathbb{N}} |x_n|.$$

Lemma 1.5.1. Let $a, b \geq 0$ and $0 \leq t \leq 1$. Then $((1-t)a + tb)^p \leq (1-t)a^p + tb^p$.

Proof.

□

Lemma 1.5.2. Let $a, b \geq 0$ and $0 \leq t \leq 1$. Then $(a+b)^p \leq 2^{p-1}(a^p + b^p)$.

Proof.

□

Lemma 1.5.3 (Minkowski's Inequality). Let $1 \leq p \leq \infty$. For any $x, y \in \ell^p$, we have $\|x + y\|_{\ell^p} \leq \|x\|_{\ell^p} + \|y\|_{\ell^p}$.

Theorem 1.5.1. For any $1 \leq p < \infty$, $(\ell^p, \|\cdot\|_{\ell^p})$ is a Banach space.

Proof. Lemma 1.5.1 establishes that ℓ^p is a vector space. Lemma 1.5.3 established that $\|\cdot\|_{\ell^p}$ satisfies the triangle inequality; the remaining norm axioms are routine to check. Hence $(\ell^p, \|\cdot\|_{\ell^p})$ is a normed linear space.

Let $\{f_n\}_{n=1}^{\infty} \subseteq \ell^p$ be a Cauchy sequence. For any $j, n, m \in \mathbb{N}$, we have:

$$\begin{aligned} |f_n(j) - f_m(j)| &\leq (|f_n(j) - f_m(j)|^p)^{\frac{1}{p}} \\ &\leq \left(\sum_{k=1}^{\infty} |f_n(k) - f_m(k)|^p \right)^{\frac{1}{p}} \\ &= \|f_n - f_m\|_{\ell^p}. \end{aligned}$$

Hence $\{f_n(j)\}_{n=1}^{\infty} \subseteq \mathbb{F}$ is Cauchy. Since $\mathbb{F}$ is complete, the limit of this sequence exists. Define $f(j) := \lim_{n \rightarrow \infty} f_n(j)$. We will show that $f \in \ell^p$ and that $f_n \rightarrow f$ in ℓ^p .

Find $N \in \mathbb{N}$ sufficiently large such that for all $n \geq N$, we have $\|f_n - f_N\|_{\ell^p} \leq 1$. Taking p^{th} powers on both sides gives the following statement: for all $n \geq N$ we have $\sum_{j=1}^{\infty} |f_n(j) - f_N(j)|^p \leq 1$. Now fix $J \in \mathbb{N}$. For all $n \geq N$ we have $\sum_{j=1}^J |f_n(j) - f_N(j)|^p \leq 1$. Take $n \rightarrow \infty$. Then we have $\sum_{j=1}^J |f(j) - f_N(j)|^p \leq 1$. In particular, for all $J \in \mathbb{N}$ we have:

$$\begin{aligned} \sum_{j=1}^J |f(j)|^p &= \sum_{j=1}^J |f(j) - f_N(j) + f_N(j)|^p \\ &\leq \sum_{j=1}^J 2^{p-1} (|f(j) - f_N(j)|^p + |f_N(j)|^p) \quad \text{Lemma 1.5.2} \\ &= 2^{p-1} \left(\sum_{j=1}^J |f(j) - f_N(j)|^p + \sum_{j=1}^J |f_N(j)|^p \right) \\ &\leq 2^{p-1} (1 + \|f_N\|_{\ell^p}^p) \\ &< \infty. \end{aligned}$$

By the Monotone Convergence Theorem, $\sum_{j=1}^{\infty} |f(j)|^p < \infty$. Hence $f \in \ell^p$.

Now let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $\|f_n - f_N\|_{\ell^p} \leq \epsilon$. Then for all $n \geq N$ and for all $j \in \mathbb{N}$, we have $\sum_{j=1}^J |f_n(j) - f_N(j)|^p \leq \epsilon^p$. Take $n \rightarrow \infty$. Then for all $J \in \mathbb{N}$ we have $\sum_{j=1}^J |f(j) - f_N(j)|^p \leq \epsilon^p$. Using the Monotone Convergence Theorem, take $J \rightarrow \infty$. Then $\sum_{j=1}^{\infty} |f(j) - f_N(j)|^p \leq \epsilon^p$. Therefore $\|f - f_N\|_{\ell^p} \leq \epsilon$. $\square$

Definition 1.5.2. The subspace of ℓ^∞ consisting of all convergent sequences is denoted $c := \{x \in \ell^\infty \mid x \text{ converges}\}$. The subspace of c consisting of all sequences which converge to zero is denoted $c_0 := \{x \in c \mid x \text{ converges to } 0\}$. The subspace of ℓ^∞ consisting of all sequences with only finitely many nonzero terms is denoted $c_{00} := \{x \in \ell^\infty \mid \text{supp}(x) < \infty\}$.

Definition 1.5.3. The standard coordinate vector in an infinite dimensional vector space is denoted $e^n := (0, \dots, 0, 1, 0, \dots)$; i.e., it is a sequence which is "1" in the n^{th} coordinate and "0" elsewhere.

Proposition 1.5.1. Let $y \in \ell^\infty$. Define $\varphi_y : \ell^1 \rightarrow \mathbb{F}$ by $\varphi_y(x) = \sum_{k=1}^{\infty} x_k y_k$. Then $\varphi \in (\ell^1)^*$ and $\|\varphi_y\|_{\ell^1}^* = \|y\|_{\ell^\infty}$.

Exercises

Exercise 1.5.1. Show that $(\ell^\infty, \|\cdot\|_\infty)$ is complete.

Proof.

$\square$

Exercise 1.5.2. Show that c and c_0 are closed in ℓ^∞ . Show that c_{00} is not closed by exhibiting a sequence in c_{00} which converges to a point $x \in \ell^\infty$ but is not in c_{00} .

Proof.

□

Exercise 1.5.3. Let $y \in \ell^\infty$ and define the linear functional:

$$\varphi_y : \ell^1 \rightarrow \mathbb{F}, \quad \varphi_y(x) := \sum_{j=1}^{\infty} x_j y_j.$$

Show that $\|\varphi_y\|_{(\ell^1)^*} = \|y\|_{\ell^\infty}$.

Proof. Note that:

$$\begin{aligned} \sum_{i=1}^{\infty} |x_i y_i| &= \sum_{i=1}^{\infty} |x_i| |y_i| \\ &\leq \|y\|_{\ell^\infty} \sum_{i=1}^{\infty} |x_i| \\ &\leq \|y\|_{\ell^\infty} \|x\|_{\ell^1}. \end{aligned}$$

Hence:

$$\begin{aligned} |\varphi_y(x)| &= \left| \sum_{i=1}^{\infty} x_i y_i \right| \\ &\leq \sum_{i=1}^{\infty} |x_i y_i| \\ &\leq \|y\|_{\ell^\infty} \|x\|_{\ell^1}. \end{aligned}$$

Dividing by $\|x\|_{\ell^1}$ and taking the supremum over all $x \in \ell^1$, $x \neq 0$ gives the inequality $\|\varphi_y\|_{(\ell^1)^*} \leq \|y\|_{\ell^\infty}$. Hence $\varphi_y \in (\ell^1)^*$.

For the converse direction, note that $e_j \in B_{\ell^1}(0, 1)$. Hence for any $j \in \mathbb{N}$, we have $|y_j| \leq |\varphi_y(e_j)| \leq \|\varphi_y\|_{(\ell^1)^*}$. Taking the supremum over all $j \in \mathbb{N}$ gives $\|y\|_{\ell^\infty} \leq \|\varphi_y\|_{(\ell^1)^*}$. Thus $\|\varphi_y\|_{(\ell^1)^*} = \|y\|_{\ell^\infty}$. □

§ 1.6. Consequences of The Baire Category Theorem

Theorem 1.6.1 (Baire Category Theorem). Let X be a complete metric space. If $\{U_n\}_{n=1}^{\infty}$ is a family of open and dense subsets of X , then $\bigcap_{n=1}^{\infty} U_n$ is still dense.

Proof.

□

Corollary 1.6.1. Let X be a complete metric space and $\{Z_n\}_{n=1}^{\infty}$ a family of closed sets. If $X = \bigcup_{n=1}^{\infty} Z_n$, then there exists some $i \in \mathbb{N}$ such that Z_i has non-empty interior.

Proof.

□

Definition 1.6.1. Let X be a metric space.

- (1) A subset Y of X is called nowhere dense if $\bar{Y}$ has empty interior.
- (2) A subset of X is called Category I if it is the countable union of nowhere dense sets.
- (3) A subset of X which is not Category I is called Category II.

Remark. Theorem 1.6.1 and Corollary 1.6.1 can be rephrased as the assertion that every complete metric space is Category II.

Theorem 1.6.2 (Banach-Steinhouse). *Let X be a Banach space and Y a normed linear space. Let $\{T_\alpha\}_{\alpha \in A} \subset \mathcal{L}(X, Y)$ be an arbitrary family of bounded linear operators. Then $\sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$ if and only if $\{x \mid \sup_{\alpha \in A} \|T_\alpha x\| < \infty\}$ is Category II*

Proof. Suppose $C := \sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$. Then for all $x \in X$ and $\alpha \in A$, we have $\|T_\alpha x\|_Y \leq \|T_\alpha\| \|x\|_X \leq C \|x\|$. Hence $\{x \mid \sup_{\alpha \in A} \|T_\alpha x\| < \infty\} = X$, which is Category II by the Baire Category Theorem.

Conversely, suppose $F := \{x \mid \sup_{\alpha \in A} \|T_\alpha x\| < \infty\}$ is Category II. Define $F_n := \{x \mid \sup_{\alpha \in A} \|T_\alpha x\| \leq n\}$. We will show that F_n is closed in X . Note that:

$$F_n = \bigcap_{\alpha \in A} \{x \mid \|T_\alpha x\| \leq n\}.$$

Note that $\|\cdot\|$ and T_α are continuous, hence the composition $g_\alpha := \|T_\alpha(\cdot)\|$ is continuous. The above expression then becomes:

$$\begin{aligned} F_n &= \bigcap_{\alpha \in A} \{x \mid \|T_\alpha x\| \leq n\} \\ &= \bigcap_{\alpha \in A} \{x \mid g_\alpha(x) \leq n\} \\ &= \bigcap_{\alpha \in A} \{x \mid g_\alpha(x) \in [0, n]\} \\ &= \bigcap_{\alpha \in A} g_\alpha^{-1}([0, n]). \end{aligned}$$

Hence F_n is closed.

Since $F = \bigcup_{n=1}^{\infty} F_n$, by Corollary 1.6.1 there exists some $n \in \mathbb{N}$ such that F_n has nonempty interior. Choose $x_0 \in X$ and $r > 0$ such that $\bar{B}(x_0, r) \subset F_n$. Let $x \in X$ such that $\|x\| \leq 1$. Fix $\alpha \in A$. Note that $x_0 + rx \in \bar{B}(x_0, r)$, hence:

$$\begin{aligned} \|T_\alpha x\|_Y &= \left\| T_\alpha \left(\frac{1}{r} (x_0 + rx - x_0) \right) \right\|_Y \\ &\leq \frac{1}{r} (\|T_\alpha(x_0 + rx)\|_Y + \|T_\alpha x_0\|_Y) \\ &\leq \frac{2n}{r} \\ &< \infty. \end{aligned}$$

Taking the supremum over all α and $\|x\| \leq 1$ gives $\sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$. $\square$

Corollary 1.6.2 (Uniform Boundedness Principle). *Let X be a Banach space and Y a normed linear space. Let $\{T_\alpha\}_{\alpha \in A} \subset \mathcal{L}(X, Y)$ be an arbitrary family of bounded linear operators. If $\sup_{\alpha \in A} \|T_\alpha x\|_Y < \infty$ for all $x \in X$, then $\sup_{\alpha \in A} \|T_\alpha\| < \infty$.*

Proof. The set $\{x \mid \sup_{\alpha \in A} \|T_\alpha x\| < \infty\} = X$ is Category II. By Theorem 1.6.2 we have that $\sup_{\alpha \in A} \|T_\alpha\| < \infty$. $\square$

Definition 1.6.2. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ linear.

- (1) The graph of T is $\text{Gr}(T) := \{(x, Tx) \mid x \in X\}$.
- (2) The map T has a closed graph if $\text{Gr}(T)$ is closed in $X \times Y$.

Remark. Regardless if $X \times Y$ is given the product or box topology, the induced metrics are equivalent.

Proposition 1.6.1. *Let X, Y be normed spaces and $T : X \rightarrow Y$ be linear. The following are equivalent:*

- (1) T has a closed graph;
- (2) If $\{x_n\}_{n=1}^\infty$ is a sequence such that $x_n \rightarrow x$ and $Tx_n \rightarrow y$, then $y = Tx$.

Proof. Suppose T has a closed graph. Let $\{x_n\}_{n=1}^\infty$ be a sequence such that $x_n \rightarrow x$ and $Tx_n \rightarrow y$. Then $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \text{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \rightarrow (x, y)$. But since T has a closed graph, $(x, y) \in \text{Gr}(T)$. Hence $y = Tx$.

Conversely, if $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \text{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \rightarrow (x, y) \in X \times Y$, then $x_n \rightarrow x$ and $Tx_n \rightarrow y$. By hypothesis $y = Tx$, hence $(x, y) \in \text{Gr}(T)$. Thus T has a closed graph. $\square$

Remark. A linear map T having a closed graph is weaker than T being continuous.

Example 1.6.1. Recall the following standard result from an introductory analysis course: let $J \subseteq \mathbb{R}$ be a bounded interval and let $\{f_n\}_{n=1}^\infty \subseteq J^\mathbb{R}$ be a sequence of functions. Suppose that there exists $x_0 \in J$ such that $\{f_n(x_0)\}_{n=1}^\infty$ converges, and that the sequence $\{f'_n\}_{n=1}^\infty$ of derivatives exists on J and converges uniformly on J to a function g . Then the sequence $\{f_n\}_{n=1}^\infty$ converges uniformly on J to a function f that has a derivative at every point of J and $f' = g$.

Indeed, define $D : C^1([0, 1]) \rightarrow C([0, 1])$ by $Df = f'$. Let $\{f_n\}_{n=1}^\infty \subseteq C^1([0, 1])$ be a sequence of functions such that $f_n \rightarrow f$ and $Df_n \rightarrow g$. Note that the convergence of $\{Df_n\}_{n=1}^\infty$ will be uniform by the compactness of $[0, 1]$. Hence by the above result, we have that $g = Df$. By Proposition 1.6.1, D has a closed graph. However, we showed in Example 1.3.3 that D is not continuous.

Definition 1.6.3. Let X, Y be normed linear spaces. A map $T : X \rightarrow Y$ is open if $T(U)$ is open in Y provided U is open in X .

Theorem 1.6.3 (Open-Mapping Theorem). *Let $T : X \rightarrow Y$ be a continuous, surjective, and linear mapping between Banach spaces. Then T is open.*

Remark. Many times throughout the following proof we will be using the fact that $B(0, n) = nB(0, 1)$.

Proof. Note that $Y = \bigcup_{n=1}^{\infty} T(nB_X(0, 1))$, hence $Y = \bigcup_{n=1}^{\infty} \overline{T(nB_X(0, 1))}$. Since Y is Banach, the Baire Category Theorem says there exists some $n \in \mathbb{N}$ such that $\overline{nT(B_X(0, 1))}$ has nonempty interior. So there exists $y_0 \in Y$ and $r > 0$ such that:

$$\begin{aligned} B_Y(y_0, r) &\subset \overline{nT(B_X(0, 1))} \\ &= \overline{nT(B_X(0, 1))^3}. \end{aligned}$$

Let $\epsilon > 0$. Since $y_0 \in B_Y(y_0, r)$, surely $y_0 \in \overline{nT(B_X(0, 1))}$. Hence there exists some $x_1 \in B_X(0, 1)$ such that $\|y_0 - nTx_1\|_Y < \epsilon$. Now let $y \in B(0, 1)$ be arbitrary. Then $y_0 + ry \in B_Y(y_0, r) \subset \overline{nT(B_X(0, 1))}$. Hence there exists $x_2 \in B_X(0, 1)$ such that $\|y_0 + ry - nTx_2\|_Y < \epsilon$. We can write:

$$\begin{aligned} y &= \frac{1}{r}(ry + y_0) - \frac{1}{r}y_0 \\ &= \frac{1}{r}(ry + y_0 - nTx_2) + \frac{n}{r}Tx_2 - \frac{1}{r}(y_0 - nTx_1) - \frac{n}{r}Tx_1 \\ &= \frac{1}{r}((y_0 + ry) - nTx_2) - \frac{1}{r}(y_0 - nTx_1) + \frac{n}{r}T(x_2 - x_1). \end{aligned}$$

We can rearrange the above equation as:

$$y - \frac{n}{r}T(x_2 - x_1) = \frac{1}{r}((y_0 + ry) - nTx_2) - \frac{1}{r}(y_0 - nTx_1).$$

Hence:

$$\begin{aligned} \left\| y - \frac{n}{r}T(x_2 - x_1) \right\| &= \left\| \frac{1}{r}((y_0 + ry) - nTx_2) - \frac{1}{r}(y_0 - nTx_1) \right\| \\ &\leq \frac{1}{r} \|((y_0 + ry) - nTx_2)\| + \frac{1}{r} \|(y_0 - nTx_1)\| \\ &< \frac{2\epsilon}{r}. \end{aligned}$$

Take $x' := x_2 - x_1 \in B(0, 2)$. Then since $\left\| y - \frac{n}{r}T(x') \right\| < \frac{2\epsilon}{r}$ can be made arbitrarily small, we have that $y \in \overline{T(B_X(0, \frac{2n}{r}))}$. Take $\lambda := \frac{2n}{r}$. Thus we've established the inclusion $B_Y(0, 1) \subset \overline{TB_X(0, \lambda)}$.

³Follows from the fact that $\overline{B(x, r)} = \overline{B}(x, r)$ and the above remark.

Let $y \in B_Y(0, 1) \subset \overline{T(B_X(0, \lambda))}$. Find $x_1 \in B_X(0, \lambda)$ such that $\|y - Tx_1\|_Y < \frac{1}{2}$. Then:

$$\begin{aligned} y - Tx_1 &\in B_Y\left(0, \frac{1}{2}\right) \\ &= \frac{1}{2}B_Y(0, 1) \\ &\subset \frac{1}{2}\overline{T(B_X(0, \lambda))} \\ &= \overline{\frac{1}{2}T(B_X(0, \lambda))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2}\right)\right)}. \end{aligned}$$

Now pick $x_2 \in B_X(0, \frac{\lambda}{2})$ such that $\|y - Tx_1 - Tx_2\| < \frac{1}{2^2}$. Then:

$$\begin{aligned} y - Tx_1 - Tx_2 &\in B_Y\left(0, \frac{1}{2^2}\right) \\ &= \frac{1}{2^2}B_Y(0, 1) \\ &\subset \frac{1}{2^2}\overline{T(B_X(0, \lambda))} \\ &= \overline{\frac{1}{2^2}T(B_X(0, 1))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2^2}\right)\right)}. \end{aligned}$$

Inductively, pick $x_k \in B_X(0, \frac{\lambda}{2^{k-1}})$ such that $\left\|y - \sum_{j=1}^k Tx_j\right\| < \frac{1}{2^k}$. Note that for all $k \in \mathbb{N}$, we have $\|x_k\|_X < \frac{\lambda}{2^{k-1}}$. Therefore $\sum_{j=1}^{\infty} x_j$ is absolutely convergent. Since X is Banach, by Proposition 1.1.2 the series $\sum_{j=1}^{\infty} x_j$ is convergent. In particular:

$$\begin{aligned} \left\|\sum_{j=1}^{\infty} x_j\right\|_X &\leq \sum_{j=1}^{\infty} \|x_j\|_X \\ &< \sum_{j=1}^{\infty} \frac{\lambda}{2^{j-1}} \\ &= 2\lambda. \end{aligned}$$

Hence $\sum_{j=1}^{\infty} x_j \in B_X(0, 2\lambda)$. Now note that $\sum_{j=1}^k x_j \rightarrow \sum_{j=1}^{\infty} x_j$ in X . Also, $\left\|y - \sum_{j=1}^k Tx_j\right\| < \frac{1}{2^k}$ implies that $T\left(\sum_{j=1}^k x_j\right) \rightarrow y$ in Y . Since T is continuous, T has a closed graph. Hence $y = T\sum_{j=1}^{\infty} x_j \in T(B_X(0, 2\lambda))$. Since $y \in B_Y(0, 1)$ was arbitrary, we've established the inclusion $B_Y(0, 1) \subset T(B_X(0, 2\lambda))$.

Now let $U \subseteq X$ be open. Let $y_0 \in T(U)$. Since T is surjective, find $x_0 \in U$ such that

$Tx_0 = y_0$. Since U is open there exists an $\epsilon > 0$ such that $B_X(x_0, \epsilon) \subseteq U$. Therefore:

$$\begin{aligned} T(U) &\supset T(B_X(x_0, \epsilon)) \\ &= Tx_0 + \frac{\epsilon}{2\lambda} T(B_X(0, 2\lambda)) \\ &\supset y_0 + \frac{\epsilon}{2\lambda} B_Y(0, 1) \\ &= B_Y\left(y_0, \frac{\epsilon}{2\lambda}\right) \end{aligned}$$

Thus $T(U)$ is open, establishing T as an open map. $\square$

Corollary 1.6.3. *Let $T : X \rightarrow Y$ be a bijective, linear, and continuous map between Banach spaces. Then T^{-1} is continuous.*

Proof. Since T is continuous, T has a closed graph. By the Open-Mapping Theorem T is an open map. Hence $(T^{-1})^{-1}(U) = T(U)$ is open. $\square$

Corollary 1.6.4. *Let X be a vector space. Suppose $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms such that $(X, \|\cdot\|_\alpha)$ and $(X, \|\cdot\|_\beta)$ are Banach. If there exists a $C > 0$ such that $\|x\|_\alpha \leq C\|x\|_\beta$ for all $x \in X$, then there exists $C' > 0$ such that $\|x\|_\beta \leq C'\|x\|_\alpha$ for all $x \in X$.*

Proof. Certainly the map $\text{id} : (X, \|\cdot\|_\beta) \rightarrow (X, \|\cdot\|_\alpha)$ is bijective and linear. Moreover, $\|\text{id} x\|_\alpha = \|x\|_\alpha \leq C\|x\|_\beta$. Hence id is continuous. By Corollary 1.6.3 id^{-1} is continuous. Hence there exists $C' > 0$ such that $\|x\|_\beta = \|\text{id}^{-1} x\|_\beta \leq C'\|x\|_\alpha$. $\square$

Remark. The above corollary can be rephrased as: if $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms such that $(X, \|\cdot\|_\alpha)$ and $(X, \|\cdot\|_\beta)$ are Banach, then $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are equivalent.

Theorem 1.6.4 (Closed Graph Theorem). *Let X and Y be Banach spaces. If $T : X \rightarrow Y$ is linear and has a closed graph, then T is continuous.*

Proof. Define $\|\cdot\|_* : X \rightarrow \mathbb{R}$ by $\|x\|_* := \|x\|_X + \|Tx\|_Y$. It is routine to check that $(X, \|\cdot\|_*)$ is a normed linear space. We will check that $(X, \|\cdot\|_*)$ is Banach. Suppose $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_*$. Then clearly $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_X$ and $\{Tx_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_Y$. Since X and Y are Banach, there exists $x \in X$ and $y \in Y$ such that $x_k \rightarrow x$ with respect to $\|\cdot\|_X$ and $Tx_k \rightarrow y$ with respect to $\|\cdot\|_Y$. Since T has a closed graph, $y = Tx$. Hence:

$$\begin{aligned} \|x - x_k\|_* &= \|x - x_k\|_X + \|Tx - Tx_k\|_Y \\ &\rightarrow 0 \text{ as } k \rightarrow \infty. \end{aligned}$$

Thus $x_k \rightarrow x$ with respect to $\|\cdot\|_*$, establishing that $(X, \|\cdot\|_*)$ is Banach. But since $(X, \|\cdot\|_X)$ and $(X, \|\cdot\|_*)$ are Banach, and since $\|x\|_X \leq \|x\|_*$, Corollary 1.6.4 says there exists $C' > 0$ such that $\|x\|_* \leq C'\|x\|_X$. Therefore $\|Tx\|_Y \leq \|x\|_* \leq C'\|x\|_X$ for all $x \in X$. Thus T is continuous. $\square$

Example 1.6.2. *jan 29th, bijective but inverse is unbounded*

Example 1.6.3. bounded but inverse is unbounded**Exercises**

Exercise 1.6.1. Let f be infinitely differentiable on $\mathbb{R}$. Suppose for any $x \in \mathbb{R}$ there is some n such that $f^{(n)}(x) = 0$ (note that n may depend on x). Show that f must be a polynomial. Hint: Consider the sets $Z_n = \{x \mid f^{(n)}(x) = 0\}$ and recall Baire's Category Theorem.

Proof.

□

Exercise 1.6.2. Let $T : X \rightarrow Y$ be a linear map between normed spaces. Prove the assertion made in class: The graph $\{(x, Tx) \mid x \in X\}$ of T is closed in $X \times Y$ if and only if:

$$\text{if } x_n \rightarrow x \text{ and } Tx_n \rightarrow y \text{ then } y = Tx.$$

Proof.

□

Exercise 1.6.3. Define $T : c_0 \rightarrow c_0$ by $T(\{x_n\}_{n=1}^\infty) = \{x_{n+1}\}_{n=1}^\infty$. Determine if T is: injective, surjective, open, closed, invertible

Proof.

- Let $(x_1, x_2, x_3, \dots)$ and $(y_1, y_2, y_3, \dots)$ be sequences in c_0 and $\alpha \in \mathbb{F}$. Observe that:

$$\begin{aligned} T((x_1, x_2, x_3, \dots) + \alpha(y_1, y_2, y_3, \dots)) &= T((x_1 + \alpha y_1, x_2 + \alpha y_2, x_3 + \alpha y_3, \dots)) \\ &= (x_2 + \alpha y_2, x_3 + \alpha y_3, x_4 + \alpha y_4, \dots) \\ &= (x_2, x_3, x_4, \dots) + \alpha(y_2, y_3, y_4, \dots) \\ &= T((x_1, x_2, x_3, \dots)) + \alpha T((y_1, y_2, y_3, \dots)) \end{aligned}$$

Thus T is linear.

- Consider the sequences $(1, 0, 0, \dots)$ and $(2, 0, 0, \dots)$. These are elements of c_0 which are not equal, but $T((1, 0, 0, \dots)) = (0, 0, 0, \dots) = T((2, 0, 0, \dots))$. Therefore T is not injective.
- Let $(x_1, x_2, x_3, \dots) \in c_0$. Then $T((0, x_1, x_2, x_3, \dots)) = (x_1, x_2, x_3, \dots)$. Thus T is surjective.
- Let $(x_1, x_2, x_3, \dots) \in c_0$. Observe that:

$$\begin{aligned} \|T((x_1, x_2, x_3, \dots))\|_{\ell^\infty} &= \|(x_2, x_3, x_4, \dots)\|_{\ell^\infty} \\ &= \sup\{|x_i| \mid i = 2, 3, 4, \dots\} \\ &\leq \sup\{|x_i| \mid i = 1, 2, 3, \dots\} \\ &= \|(x_1, x_2, x_3, \dots)\|_{\ell^\infty}. \end{aligned}$$

Thus T is bounded, hence continuous.

- Since T is surjective, continuous, and linear, by the Open-Mapping Theorem T is open.

- Since T is continuous, T has a closed graph.
- Since T is not bijective, T is not invertible.

□

Exercise 1.6.4. Let $L : X \rightarrow Y$ be a linear map between Banach spaces. Suppose that the conditions $x_n \rightarrow 0$ and $Lx_n \rightarrow y$ imply that $y = 0$. Prove that L is continuous.

Proof. Let $\{(x_n, Tx_n)\}_{n=1}^{\infty} \subseteq \text{Gr}(L)$ be a sequence such that $(x_n, Tx_n) \rightarrow (x, y) \in X \times Y$. Then $x_n \rightarrow x$ and $Tx_n \rightarrow y$. Define $u_n := x_n - x$. Then $u_n \rightarrow 0$ and $Tu_n \rightarrow y - Tx$. By hypothesis $y - Tx = 0$, giving $y = Tx$. Thus $(x, y) \in \text{Gr}(T)$, which means T has a closed graph. By the Closed Graph Theorem, T is continuous. □

Exercise 1.6.5. Prove that if a closed map has an inverse then the inverse is also closed.

Proof. Let $\{y_n\}_{n=1}^{\infty}$ be a sequence in Y such that $y_n \rightarrow y$ and $T^{-1}y_n \rightarrow x$. Since T admits an inverse, it is bijective. Hence for each $y_n \in Y$, there exists $x_n \in X$ such that $y_n = Tx_n$. The two sequences above can be rewritten as $Tx_n \rightarrow y$ and $T^{-1}(Tx_n) = x_n \rightarrow x$. Since T has a closed graph, $y = Tx$. Since T is bijective, $x = T^{-1}y$. Thus T^{-1} has a closed graph. □

Exercise 1.6.6. Let $L : X \rightarrow Y$ be a continuous linear surjection, where X and Y are Banach spaces. Let $y_n \rightarrow y$ in Y . Prove that there exists points $x_n \in X$ and a constant $c \in \mathbb{R}$ such that $Lx_n = y_n$, the sequence $\{x_n\}_{n=1}^{\infty}$ converges, and $\|x_n\| \leq c\|y_n\|$.

Proof.

□

§ 1.7. Continuous Dual, Adjoint Maps, and Weak Convergence

Definition 1.7.1. Let X and Y be normed linear spaces and $T \in \mathcal{L}(X, Y)$. The adjoint of T is a map $T^* : Y^* \rightarrow X^*$ given by $T^*(\varphi) = \varphi(Tx)$.

Proposition 1.7.1. Let X and Y be normed linear spaces and $T \in \mathcal{L}(X, Y)$. Then $T^* \in \mathcal{L}(Y^*, X^*)$, and $\|T^*\|_{\mathcal{L}(Y^*, X^*)} = \|T\|_{\mathcal{L}(X, Y)}$.

Proof.

□

Exercises

Exercise 1.7.1. Let X be a normed linear space, let $K \subset X$, and let $\Omega \subset X^*$. Show that $K^\perp$ is closed in X^* and that $\Omega_\perp$ is closed in X .

Proof. Let $\{\varphi_n\}_{n=1}^\infty \subset K^\perp$ be a sequence such that $\varphi_n \rightarrow \Phi \in X^*$. Let $k \in K$ be arbitrary. Then $\varphi_n(k) \rightarrow \Phi(k)$. Since $\varphi_n(k) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\Phi(k) = 0$. Since k was arbitrary, we have $\Phi \in K^\perp$. Since $K^\perp$ contains all of its limit points, it is closed in X^* .

Let $\{x_n\}_{n=1}^\infty \subset \Omega_\perp$ be a sequence such that $x_n \rightarrow y \in X$. Let $\varphi \in \Omega$ be arbitrary. Then $\varphi(x_n) \rightarrow \varphi(y)$ since φ is continuous. Since $\varphi(x_n) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\varphi(y) = 0$. Since φ was arbitrary, $y \in \Omega_\perp$. Since $\Omega_\perp$ contains all of its limit points, it is closed in X . $\square$

Exercise 1.7.2. Let X be a normed linear space and $K \subset X$. Show that $(K^\perp)_\perp$ is the closure of the linear space K in X .

Proof.

$\square$

§ 1.8. L^p Spaces

Exercises

Exercise 1.8.1. For which $r \in \mathbb{R}$ is the function $f(x) = \frac{1}{|x|^r}$ in $L^1([0, 1])$? What about $L^p([0, 1])$?

Proof. Since $x \in [0, 1]$, we have $|x| = x$. Note that:

$$\int_0^1 x^{-r} dx = -\frac{1}{(r-1)x^{r-1}} \Big|_0^1$$

The right-hand side of this equation can only be evaluated if $r-1 > 0$. Hence $\frac{1}{|x|^r} \in L^1([0, 1])$ provided $r > 1$. Similarly, note that:

$$\begin{aligned} \int_0^1 (x^{-r})^p dx &= \int_0^1 x^{-rp} dx \\ &= -\frac{1}{(rp-1)x^{rp-1}} \Big|_0^1 \end{aligned}$$

The right-hand side of this equation can only be evaluated if $rp-1 > 0$. Hence $\frac{1}{|x|^r} \in L^p([0, 1])$ provided $r > \frac{1}{p}$. $\square$

Chapter 2

Hilbert Spaces

§ 2.1. Inner Product Spaces

Definition 2.1.1. Let V be a vector space over $\mathbb{F}$. An *inner product* is a map:

$$(\cdot, \cdot) : V \rightarrow \mathbb{F}$$

satisfying for all $x, y \in V$:

- (1) $(x, x) \geq 0$ and $(x, x) = 0$ if and only if $x = 0$;
- (2) $(\cdot, \cdot)$ is linear in the first slot.
- (3) $(x, y) = \overline{(y, x)}$.

Remark. Note that an inner product over $\mathbb{R}$ (resp. $\mathbb{C}$) can be defined as a positive-definite and symmetric (resp. Hermitian) bilinear form (resp. sesquilinear form). From (2) and (3) we have:

$$\begin{aligned} (x, \lambda y) &= \overline{(\lambda y, x)} \\ &= \overline{\lambda(y, x)} \\ &= \bar{\lambda} \overline{(y, x)} \\ &= \bar{\lambda} (x, y). \end{aligned}$$

Definition 2.1.2. Let V be an inner product space. We say $x, y \in V$ are orthogonal and write $x \perp y$ if $(x, y) = 0$.

Example 2.1.1.

- (1) The space $\mathbb{C}^n$ with $(x, y) := \sum_{j=1}^n x_j \overline{y_j}$ is an inner product space.
- (2) The space ℓ^2 with $(x, y) := \sum_{j=1}^{\infty} x_j \overline{y_j}$ is an inner product space. Note that the normed induced by the inner product is precisely the familiar ℓ^2 -norm.
- (3) The space $L^2(\Omega)$ with $(f, g) := \int_{\Omega} f(x) \overline{g(x)} dx$ is an inner product space. The norm induced by the inner product is also the familiar L^2 -norm.

Proposition 2.1.1. Let X be an inner product space and $x, y \in X, y \neq 0$. There exists $c \in \mathbb{F}$ such that $(x - cy) \perp y$.

Proof. By definition $(x - cy) \perp y$ means $(x - cy, y) = 0$. Hence $(x, y) - c(y, y) = 0$. Solving for c gives $c = \frac{(x, y)}{\|y\|^2}$. □

Theorem 2.1.1. Let X be an inner product space. Let $x, y \in X$.

- (1) If $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ (Pythagorean Theorem).
 (2) $|(x, y)| \leq \|x\| \|y\|$ (Cauchy-Schwartz Inequality).
 (3) $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (Parallelogram law).

Proof. (1) Observe that:

$$\begin{aligned}
 \|x + y\|^2 &= (x + y, x + y) \\
 &= (x, x) + (x, y) + (y, x) + (y, y) \\
 &= (x, x) + (x, y) + \overline{(x, y)} + (y, y) \\
 &= (x, x) + 2\Re((x, y)) + (y, y) \\
 &= (x, x) + (y, y) \\
 &= \|x\|^2 + \|y\|^2.
 \end{aligned}$$

(2) Assume $y \neq 0$. Find c such that $(x - cy) \perp y$. Then:

$$\begin{aligned}
 \|x\|^2 &= \|(x - cy) + cy\|^2 \\
 &= \|(x - cy)\|^2 + \|cy\|^2 \\
 &\geq |c|^2 \|y\|^2.
 \end{aligned}$$

Note that $c = \frac{(x, y)}{\|y\|^2}$ by Proposition 2.1.1. Hence $\|x\|^2 \geq \frac{|(x, y)|^2}{\|y\|^4} \|y\|^2$. Solving for $|(x, y)|$ establishes the inequality.

(3) Exercise 2.1.1. □

Proposition 2.1.2. Let V be an inner product space. Define $\|\cdot\| : V \rightarrow \mathbb{F}$ by $\|x\| = \sqrt{(x, x)}$. Then $(V, \|\cdot\|)$ is a normed linear space.

Proof. Let $v, w \in V$ and $\alpha \in \mathbb{F}$. We have that:

$$\begin{aligned}
 \|\alpha v\| &= \sqrt{(\alpha v, \alpha v)} \\
 &= \sqrt{\alpha \bar{\alpha} (v, v)} \\
 &= \sqrt{|\alpha|^2 (v, v)} \\
 &= |\alpha| \|v\|.
 \end{aligned}$$

Thus $\|\cdot\|$ is homogeneous. By Cauchy-Schwartz:

$$\begin{aligned}
 \|v + w\|_2^2 &= \langle v + w, v + w \rangle \\
 &= \langle v, v \rangle + \langle v, w \rangle + \langle w, v \rangle + \langle w, w \rangle \\
 &= \|v\|_2^2 + \langle v, w \rangle + \overline{\langle v, w \rangle} + \|w\|_2^2 \\
 &= \|v\|_2^2 + 2\Re(\langle v, w \rangle) + \|w\|_2^2 \\
 &\leq \|v\|_2^2 + 2|\langle v, w \rangle| + \|w\|_2^2 \\
 &\leq \|v\|_2^2 + 2\|v\|_2 \|w\|_2 + \|w\|_2^2 \\
 &= (\|v\|_2 + \|w\|_2)^2,
 \end{aligned}$$

Squaring both sides establishes the triangle inequality. For positive-definiteness, suppose $\|v\| = 0$. Then $\sqrt{(v, v)} = 0$, so it must be the case that $v = 0$. $\square$

Definition 2.1.3. Let X be an inner product space and $A \subset X$ a subspace. The orthogonal complement of A is the set $A^\perp := \{x \in X \mid (x, a) = 0 \text{ for all } a \in A\}$.

Proposition 2.1.3. If X is an inner product space and $A \subset X$ is a subspace, then $A^\perp$ is closed in X .

Proof. Define $\varphi_a : X \rightarrow \mathbb{F}$ by $x \mapsto (x, a)$. This map is continuous. Then $A^\perp = \bigcap_{a \in A} \varphi_a^{-1}(\{0\})$. Since φ_a is continuous, $A^\perp$ is closed. $\square$

Exercises

Exercise 2.1.1. Prove the Parallelogram Law in any inner-product space.

Proof. Observe that:

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= (x + y, x + y) + (x - y, x - y) \\ &= (x, x) + 2\Re((x, y)) + (y, y) + (x, x) - 2\Re((x, y)) + (y, y) \\ &= 2(x, x) + 2(y, y) \\ &= 2\|x\|^2 + 2\|y\|^2. \end{aligned} \quad \square$$

Exercise 2.1.2. Let X be an inner-product space over $\mathbb{C}$ and fix $y \in X$. Prove that the map $\varphi_y : X \rightarrow \mathbb{C}$ defined by $\varphi_y(x) = (x, y)$ is continuous.

Proof. Let $x_1, x_2 \in X$. Observe that:

$$\begin{aligned} |\varphi_y(x_1) - \varphi_y(x_2)| &= |(x_1, y) - (x_2, y)| \\ &= |(x_1 - x_2, y)| \\ &\leq \|x_1 - x_2\| \|y\|. \end{aligned}$$

Since φ_y is Lipschitz, φ_y is continuous. $\square$

Exercise 2.1.3. Prove that in an inner-product space X , if $\{x_n\}_{n=1}^\infty \subset X$ is a sequence and $y \in X$ satisfy:

$$\|x_n\| \rightarrow \|y\| \text{ and } (x_n, y) \rightarrow \|y\|^2,$$

then $x_n \rightarrow y$.

Proof. Observe that:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \|x_n - y\|^2 &= \lim_{n \rightarrow \infty} (x_n - y, x_n - y) \\
 &= \lim_{n \rightarrow \infty} \left((x_n, x_n) - (y, x_n) - (x_n, y) + (y, y) \right) \\
 &= \lim_{n \rightarrow \infty} \left(\|x_n\|^2 - 2\Re((x_n, y)) + \|y\|^2 \right) \\
 &= \lim_{n \rightarrow \infty} \|x_n\|^2 - \lim_{n \rightarrow \infty} 2\Re((x_n, y)) + \lim_{n \rightarrow \infty} \|y\|^2 \\
 &= \lim_{n \rightarrow \infty} \|x_n\|^2 - 2\Re\left(\lim_{n \rightarrow \infty} (x_n, y)\right) + \lim_{n \rightarrow \infty} \|y\|^2 \\
 &= \|y\|^2 - 2\|y\|^2 + \|y\|^2 \\
 &= 0.
 \end{aligned}$$

Since $\|x_n - y\|^2 \rightarrow 0$, it must be the case that $\|x_n - y\| \rightarrow 0$. Thus $x_n \rightarrow y$. $\square$

§ 2.2. Hilbert Spaces

Definition 2.2.1. An inner product space which is complete is called a Hilbert space.

Remark. It is standard to assume that the base field of a Hilbert space is $\mathbb{C}$.

Theorem 2.2.1. Let $\mathcal{H}$ be a Hilbert space and $K \subset \mathcal{H}$ a nonempty subset which is convex and closed. For any $x \in \mathcal{H}$, there exists a unique $y \in K$ which is closed to x .

Proof. Let $x \in X$. Let $\alpha := \text{dist}(x, K)$. Find a sequence $\{y_n\}_{n=1}^\infty \subset K$ such that $\|x - y_n\| \rightarrow \alpha$ as $n \rightarrow \infty$. For any $n, m \in \mathbb{N}$:

$$\begin{aligned}
 \|y_n - y_m\|^2 &= \|(y_n - x) + (x - y_m)\|^2 \\
 &= 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - \|y_n + y_m - 2x\|^2 && \text{Parallelogram Law} \\
 &= 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - 4\left\|\frac{y_n + y_m}{2} - x\right\|^2 && \text{See footnote}^1 \\
 &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\alpha^2.
 \end{aligned}$$

Since $\|x - y_n\| \rightarrow \alpha^2$ and $\|x - y_m\| \rightarrow \alpha^2$, note that $\|y_n - y_m\|^2$ can be made arbitrary small. Hence $\{y_n\}_{n=1}^\infty$ is Cauchy. Since K is a closed subset of a Hilbert space, it is complete. Hence there exists some $y \in K$ such that $y_n \rightarrow y$. Finally:

$$\begin{aligned}
 \|x - y\| &= \left\| x - \lim_{n \rightarrow \infty} y_n \right\| \\
 &= \lim_{n \rightarrow \infty} \|x - y_n\| \\
 &= \alpha.
 \end{aligned}$$

¹Note $\frac{y_n + y_m}{2} \in K$ because K is convex.

Now if y' is any other element in K with $\|y' - x\| = \alpha$, by our previous calculation:

$$\begin{aligned}\|y - y'\|^2 &= 2\|y - x\|^2 + 2\|y' - x\|^2 - \|y + y' - 2x\|^2 \\ &\leq 2\alpha^2 + 2\alpha^2 - 4\alpha^2 \\ &= 0.\end{aligned}$$

Thus $y = y'$. □

Theorem 2.2.2. *Let $\mathcal{H}$ be a Hilbert space. Let $Z \subset \mathcal{H}$ be a closed subspace. Then $\mathcal{H} = Z \oplus Z^\perp$.*

Proof. If $z \in Z \cap Z^\perp$, then $z \perp z$; i.e., $(z, z) = 0$. Therefore $z = 0$.

Now let $x \in \mathcal{H}$. By Theorem 2.2.1, let $z \in Z$ be the unique element closest to x . We show that $x - z \in Z^\perp$. Let $z' \in Z$ be nonzero. By Proposition 2.1.1, choose $\lambda \in \mathbb{C}$ such that $\lambda z' \perp (x - z) - \lambda z'$. Write:

$$x - z = \lambda z' + (x - z) - \lambda z'.$$

Then the Pythagorean Theorem gives:

$$\begin{aligned}\|x - z\|^2 &= \|\lambda z'\|^2 + \|x - (z + \lambda z')\|^2 \\ &\geq \|\lambda z'\|^2 + \|x - z\|^2.\end{aligned}$$

Note that the inequality is a result from the fact that z is the closest element in Z to x . Subtracting $\|x - z\|^2$ from both sides yields:

$$0 \geq \|\lambda z'\|^2$$

Because $z' \neq 0$, this forces $\lambda = 0$. Since $\lambda = \frac{(x - z, z')}{\|z'\|^2}$, we obtain $(x - z, z') = 0$. As $z' \in Z$ was arbitrary, $x - z \in Z^\perp$. Consequently, $x = z + (x - z) \in Z + Z^\perp$, establishing $x \in Z \oplus Z^\perp$. □

Definition 2.2.2. Let X be an inner product space. Let $\mathcal{U} := \{u_\alpha \mid \alpha \in I\} \subset X$. We say $\mathcal{U}$ is orthogonal if for all $\alpha \neq \beta$, we have $(u_\alpha, u_\beta) = 0$. We say $\mathcal{U}$ is orthonormal if for all $\alpha, \beta \in I$, we have:

$$(u_\alpha, u_\beta) = \begin{cases} 1, & \alpha = \beta \\ 0, & \alpha \neq \beta. \end{cases}$$

Theorem 2.2.3. *Let X be a nonzero inner product space. There exists a maximal orthonormal set by inclusion*

Proof. Order all possible orthonormal sets by inclusion. Show that every chain admits an upper bound. Then apply Zorn's Lemma. □

Theorem 2.2.4. *Let X be a separable inner product space. Then any orthonormal subset is at most countable.*

Proof. Let $\{a_j\}_{j=1}^\infty$ be a countable dense subset of X . Suppose towards contradiction $\mathcal{U} := \{u_\alpha \mid \alpha \in I\}$ is an uncountable orthonormal set in X . Pick $\alpha, \beta \in I$ such that $\alpha \neq \beta$. Then:

$$\begin{aligned}\|u_\alpha - u_\beta\|^2 &= \|u_\alpha\|^2 + \|u_\beta\|^2 \\ &= 2.\end{aligned}$$

Therefore, any two elements of $\mathcal{U}$ are $\sqrt{2}$ apart. Now for all $\alpha \in I$, consider the open balls $B(u_\alpha, \frac{\sqrt{2}}{2})$. Since each open ball has diameter less than $\sqrt{2}$, each ball can contain at most one element of $\{a_j\}_{j=1}^\infty$. Moreover, the collection $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$ is disjoint. Indeed, if $y \in B(u_\alpha, \frac{\sqrt{2}}{2}) \cap B(u_\beta, \frac{\sqrt{2}}{2})$, then $\|u_\alpha - u_\beta\| \leq \|u_\alpha - y\| + \|y - u_\beta\| < \sqrt{2}$, which is a contradiction. Since $\{a_j\}_{j=1}^\infty$ is a dense subset, each a_j must be contained in a ball. We can define a surjection from $\{a_j\}_{j=1}^\infty$ to $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$. But since $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$ is indexed by an uncountable set, it must be the case that $\{a_j\}_{j=1}^\infty$ is uncountable. This is a contradiction, whence any orthonormal subset must be at most countable. $\square$

Theorem 2.2.5. *Let X be an inner product space. Let $Y \subset X$ be a subspace with $\dim Y = n$.*

- (1) *Y has an orthonormal basis $\{u_1, \dots, u_n\}$;*
- (2) *For each $y \in Y$, we can write $y = \sum_{j=1}^n (y, u_j) u_j$. Moreover, $\|y\|^2 = \sum_{j=1}^n |(y, u_j)|^2$.*
- (3) *There is an isometric isomorphism $Y \rightarrow \mathbb{C}^n$.*
- (4) *For any $x \in X$, we have $x - \sum_{j=1}^n (x, u_j) u_j \perp Y$. Moreover, $\sum_{j=1}^n (x, u_j) u_j$ is the unique element in Y closest to x ,*

Proof. (1) Apply the Gram-Schmidt process to a basis of Y .

(2) Let $y \in Y$. We can uniquely write $y = \sum_{j=1}^n a_j u_j$ for some $a_1, \dots, a_n \in \mathbb{C}$. Then $(y, u_j) = (\sum_{k=1}^n a_k u_k, u_j) = \sum_{k=1}^n a_k (u_k, u_j) = a_j$. Hence $y = \sum_{j=1}^n (y, u_j) u_j$.

(3) For any $y \in Y$, we can write $y = \sum_{j=1}^n a_j u_j$. Then $Y \rightarrow \mathbb{C}^n$ given by $y \mapsto (a_1, \dots, a_n)$ is linear, bijective, and norm preserving.

(4) Let $x \in X$. Define $a_j := (x, u_j)$ for each $j = 1, \dots, n$. For each $k = 1, \dots, n$:

$$\begin{aligned}\left(x - \sum_{j=1}^n a_j u_j, u_k\right) &= (x, u_k) - \left(\sum_{j=1}^n a_j u_j, u_k\right) \\ &= (x, u_k) - \sum_{j=1}^n a_j (u_j, u_k) \\ &= (x, u_k) - a_k \\ &= (x, u_k) - (x, u_k) \\ &= 0.\end{aligned}$$

Since any $y \in Y$ can be uniquely written as a linear combination of $u_1, \dots, u_n$, the above string of equalities implies $x - \sum_{j=1}^n (x, u_j) u_j \perp Y$. Now let $\sum_{j=1}^n b_j u_j$ be any element in Y .

Then:

$$\begin{aligned}
 \left\| x - \sum_{j=1}^n b_j u_j \right\|^2 &= \left\| x - \sum_{j=1}^n a_j u_j + \sum_{j=1}^n (a_j - b_j) u_j \right\|^2 \\
 &= \left\| x - \sum_{j=1}^n a_j u_j \right\|^2 + \left\| \sum_{j=1}^n (a_j - b_j) u_j \right\|^2 \\
 &= \left\| x - \sum_{j=1}^n a_j u_j \right\|^2 + \sum_{j=1}^n |a_j - b_j|^2 \\
 &\geq \left\| x - \sum_{j=1}^n a_j u_j \right\|^2,
 \end{aligned}$$

where we used the fact that $x - \sum_{j=1}^n a_j u_j \perp \sum_{j=1}^n (a_j - b_j) u_j$. This establishes that $\sum_{j=1}^n a_j u_j$ is the unique element in Y closest to x , where we have equality only when $a_j = b_j$ for all $j = 1, \dots, n$. $\square$

Theorem 2.2.6. *Let $\mathcal{H}$ be a Hilbert space. Let $\{x_j\}_{j=1}^\infty$ be an orthogonal sequence in $\mathcal{H}$. Then $\sum_{j=1}^\infty x_j$ converges in $\mathcal{H}$ if and only if $\sum_{j=1}^\infty \|x_j\|^2 < \infty$.*

Proof. Define:

$$\begin{aligned}
 z_n &:= \sum_{j=1}^n x_j, \\
 s_n &:= \sum_{j=1}^n \|x_j\|^2.
 \end{aligned}$$

Now $\sum_{j=1}^\infty x_j$ converges if and only if $\{z_n\}_{n=1}^\infty$ is Cauchy. For $n > m$:

$$\begin{aligned}
 \|z_n - z_m\|^2 &= \left\| \sum_{j=m+1}^n x_j \right\|^2 \\
 &= \sum_{j=m+1}^n \|x_j\|^2 \\
 &= |s_n - s_m|.
 \end{aligned}$$

From this, $\{z_n\}_{n=1}^\infty$ is Cauchy in $\mathcal{H}$ if and only if $\{s_n\}_{n=1}^\infty$ is Cauchy in $\mathbb{C}$. $\square$

Theorem 2.2.7 (Bessel's Inequality). *Let X be an inner-product space. Let $\{u_j\}_{j=1}^\infty$ be an orthonormal sequence in X . For all $x \in X$, we have $\sum_{j=1}^\infty |(x, u_j)|^2 \leq \|x\|^2$.*

Proof. For each $n \in \mathbb{N}$, define $Y_n := \text{span}\{u_1, \dots, u_n\}$. Let P_{Y_n} denote the projection onto Y_n . Then:

$$\begin{aligned} \sum_{j=1}^n |(x, u_j)|^2 &= \sum_{j=1}^n \|(x, u_j)u_j\|^2 \\ &= \left\| \sum_{j=1}^n (x, u_j)u_j \right\|^2 \\ &= \|P_{Y_n}(x)\|^2 \\ &\leq \|P_{Y_n}\|^2 \|x\|^2 \\ &= \|x\|^2. \end{aligned}$$

Note that $\|P_{Y_n}\| = 1$ is established in Exercise 2.2.5. Since n was arbitrary, by the Monotone Convergence Theorem $\sum_{j=1}^{\infty} |(x, u_j)|^2 \leq \|x\|^2$. $\square$

Theorem 2.2.8. Let $\mathcal{H}$ be a Hilbert space. Let $\{u_j\}_{j=1}^{\infty}$ be an orthonormal sequence in $\mathcal{H}$. The following are equivalent:

- (1) $\{u_j\}_{j=1}^{\infty}$ is a maximal orthonormal set;
- (2) For all $x \in \mathcal{H}$, if $x \perp \{u_j\}_{j=1}^{\infty}$, then $x = 0$;
- (3) For all $x \in \mathcal{H}$, there exists a unique sequence $\{a_j\}_{j=1}^{\infty}$ such that $x = \sum_{j=1}^{\infty} a_j u_j$.

Proof. We will only show (2) if and only if (3), as (1) if and only if (2) follows immediately by applying definitions.

Suppose for all $x \in \mathcal{H}$ we have $x \perp \{u_j\}_{j=1}^{\infty}$ implies $x = 0$. Let $x \in \mathcal{H}$. Claim: $x - \sum_{j=1}^{\infty} (x, u_j)u_j \perp \{u_j\}_{j=1}^{\infty}$. Indeed, for any $u_k \in \{u_j\}_{j=1}^{\infty}$:

$$\begin{aligned} \left(x - \sum_{j=1}^{\infty} (x, u_j)u_j, u_k \right) &= \lim_{n \rightarrow \infty} \left(x - \sum_{j=1}^n (x, u_j)u_j, u_k \right) \\ &= \lim_{n \rightarrow \infty} \left((x, u_k) - \sum_{j=1}^n ((x, u_j)u_j, u_k) \right) \\ &= \lim_{n \rightarrow \infty} ((x, u_k) - (x, u_k)) \\ &= 0. \end{aligned}$$

Hence $x - \sum_{j=1}^{\infty} (x, u_j)u_j = 0$, and rearranging gives $x = \sum_{j=1}^{\infty} (x, u_j)u_j$. Now suppose there exists a sequence $\{b_n\}_{n=1}^{\infty}$ such that $x = \sum_{j=1}^{\infty} b_j u_j$. Then $0 = \sum_{j=1}^{\infty} ((x, u_j) - b_j)u_j$, which by Bessel's Inequality results in $\sum_{j=1}^{\infty} |(x, u_j) - b_j|^2 \leq 0$. It must be the case that $(x, u_j) = b_j$ for all $j \in \mathbb{N}$, establishing uniqueness.

Let $x \in \mathcal{H}$ and suppose $x \perp \{u_j\}_{j=1}^{\infty}$. Write $x = \sum_{j=1}^{\infty} (x, u_j)u_j$. Then for all $k \in \mathbb{N}$:

$$\begin{aligned} (x, u_k) &= \left(\sum_{j=1}^{\infty} (x, u_j)u_j, u_k \right) \\ &= 0. \end{aligned}$$

Thus $x = 0$. □

Definition 2.2.3. Let $\mathcal{H}$ be a Hilbert space and $\{u_j\}_{j=1}^{\infty}$ an orthonormal sequence in $\mathcal{H}$. If any of the three equivalent statements in Theorem 2.2.8 hold true, we say $\{u_j\}_{j=1}^{\infty}$ is a orthonormal basis (or Hilbert basis).

Definition 2.2.4. Let $\mathcal{H}$ be a Hilbert space and $\{u_j\}_{j=1}^{\infty}$ an orthonormal basis. Let $x \in \mathcal{H}$.

- (1) The sequence $\{(x, u_j)\}_{j=1}^{\infty}$ is referred to as the generalized Fourier coefficients of x .
- (2) The series $\sum_{j=1}^{\infty} (x, u_j) u_j$ is called the generalized Fourier series of x .

Example 2.2.1. The set $\{e^n\}_{n=1}^{\infty}$ is an orthonormal basis in ℓ^2 . For any separable infinite dimensional hilbert space, let $\{u_j\}_{j=1}^{\infty}$ be an orthonormal basis. Define $H \rightarrow \ell^2$ by $\sum_{j=1}^{\infty} a_j u_j \mapsto (a_1, a_2, \dots)$. This is an isometric isomorphism.

Feb 12, stuff on L^2

Theorem 2.2.9 (Riez Representation Theorem). Let $\mathcal{H}$ be a Hilbert space. Let $\varphi \in \mathcal{H}^*$. There exists a unique $y \in H$ such that for all $x \in \mathcal{H}$, $\varphi(x) = (x, y)$.

Proof. If $y, y' \in \mathcal{H}$ satisfy $\varphi(x) = (x, y) = (x, y')$ for all $x \in \mathcal{H}$, then $(x, y - y') = 0$ for all $x \in \mathcal{H}$. Note that $y - y' \in \mathcal{H}$, hence we must have that $(y - y', y - y') = 0$. Thus $\|y - y'\| = 0$, hence $y = y'$.

We now prove existence. Let $\varphi \in \mathcal{H}^*$. Let $z \in N(\varphi)^\perp$ such that $\|z\| = 1$. For any $x \in H$, we have $x - \frac{\varphi(x)}{\varphi(z)} z \in N(\varphi)$ and the equation:

$$x = \left(x - \frac{\varphi(x)}{\varphi(z)} z \right) + \left(\frac{\varphi(x)}{\varphi(z)} z \right).$$

Now:

$$\begin{aligned} (x, \lambda z) &= \left(x - \frac{\varphi(x)}{\varphi(z)} z + \frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) \\ &= \left(x - \frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) + \left(\frac{\varphi(x)}{\varphi(z)} z, \lambda z \right). \end{aligned}$$

Because $z \in N(\varphi)^\perp$ and $x - \frac{\varphi(x)}{\varphi(z)} z \in N(\varphi)$, then $\left(x - \frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) = 0$. We now have:

$$\begin{aligned} (x, \lambda z) &= \left(x - \frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) + \left(\frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) \\ &= \left(\frac{\varphi(x)}{\varphi(z)} z, \lambda z \right) \\ &= \frac{\varphi(x)}{\varphi(z)} \bar{\lambda} (z, z) \\ &= \frac{\varphi(x)}{\varphi(z)} \bar{\lambda} \|z\|^2 \\ &= \frac{\varphi(x)}{\varphi(z)} \bar{\lambda}. \end{aligned}$$

Let $\lambda := \overline{\varphi(z)}$. Then $(x, \varphi(z)) = \varphi(x)$, establishing the theorem. $\square$

self-adjoint operators

Exercises

Exercise 2.2.1. Let $H = L^2([0, 1])$. Use the Gram-Schmidt orthogonalization process from linear algebra to produce an orthonormal set $\{u_1, u_2, u_3\}$ which spans the same space as $\text{span}\{1, x, x^2\}$.

Proof. Let $v_1 := 1$, $v_2 := x$, and $v_3 := x^2$. The Gram-Schmidt process gives:

$$\begin{aligned} r_1 &= v_1 = 1, \\ \|r_1\| &= \sqrt{(r_1, r_1)} = 1. \\ r_2 &= v_2 - \frac{(v_2, r_1)}{(r_1, r_1)} r_1 = x - \frac{1}{2}, \\ \|r_2\| &= \sqrt{(r_2, r_2)} = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \frac{1}{\sqrt{12}}, \\ r_3 &= v_3 - \frac{(v_3, r_1)}{(r_1, r_1)} r_1 - \frac{(v_3, r_2)}{(r_2, r_2)} r_2 = x^2 - x + \frac{1}{6}, \\ \|r_3\| &= \sqrt{(r_3, r_3)} = \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx} = \frac{1}{\sqrt{180}}. \end{aligned}$$

Thus an orthonormal basis for $\text{span}\{1, x, x^2\}$ is:

$$\begin{aligned} u_1 &:= \frac{r_1}{\|r_1\|} = 1 \\ u_2 &:= \frac{r_2}{\|r_2\|} = \sqrt{12} \left(x - \frac{1}{2}\right) \\ u_3 &:= \frac{r_3}{\|r_3\|} = \sqrt{180} \left(x^2 - x + \frac{1}{6}\right). \end{aligned} \quad \square$$

Exercise 2.2.2. Show directly that the functions:

$$u_n(x) = e^{i2\pi n x}, \quad n \in \mathbb{Z}$$

form an orthonormal sequence in $L^1([0, 1])$. Then write the function $f(x) = x$ in terms of this basis, that is find a_n such that:

$$x \sim \sum_{n=-\infty}^{\infty} a_n u_n(x).$$

What does Parseval's identity say in this case? (After some algebra you should be able to use your summation to find a formula for $\sum_{n=1}^{\infty} \frac{1}{n^2}$.)

Proof. Observe that:

$$\begin{aligned}
 \int_0^1 e^{i2\pi nx} e^{-i2\pi mx} dx &= \int_0^1 1 dx = 1. \\
 \int_0^1 e^{i2\pi nx} e^{-i2\pi mx} dx &= \int_0^1 e^{i2\pi(n-m)x} dx \\
 &= \frac{e^{i2\pi(n-m)x}}{i2\pi(n-m)} \Big|_0^1 \\
 &= \frac{e^{i2\pi(n-m)} - 1}{i2\pi(n-m)} \\
 &= \frac{\cos(2\pi(n-m)) - i \sin(2\pi(n-m)) - 1}{i2\pi(n-m)} \\
 &= \frac{1 - 0 - 1}{i2\pi(n-m)} \\
 &= 0.
 \end{aligned}$$

Thus $\{u_n\}_{n=1}^\infty$ is an orthonormal sequence in $L^1([0, 1])$. We proceed by cases on computing the Fourier coefficients of x . For $n = 0$:

$$a_0 = \int_0^1 x dx = \frac{1}{2}.$$

For $n \neq 0$:

$$\begin{aligned}
 a_n &= (x, u_n) \\
 &= \int_0^1 x e^{-i2\pi nx} dx \\
 &= -\frac{e^{i2\pi n}(e^{i2\pi n} - i2\pi n - 1)}{4n^2\pi^2} \quad \text{From Mathematica} \\
 &= \frac{i}{2n\pi}. \quad \text{Also from Mathematica}
 \end{aligned}$$

Using Parseval's Identity:

$$\begin{aligned}
 \frac{1}{3} &= \int_0^1 x^2 dx \\
 &= \|x\|^2 \\
 &= \sum_{n=-\infty}^{\infty} |a_n|^2 \\
 &= |a_0|^2 + 2 \sum_{n=1}^{\infty} |a_n|^2 \\
 &= \frac{1}{4} + 2 \sum_{n=1}^{\infty} \left| \frac{i}{2\pi n} \right|^2 \\
 &= \frac{1}{4} + \frac{1}{2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}.
 \end{aligned}$$

Solving for $\sum_{n=1}^{\infty} \frac{1}{n^2}$ gives:

$$\begin{aligned}
 \sum_{n=1}^{\infty} \frac{1}{n^2} &= 2\pi^2 \left(\frac{1}{3} - \frac{1}{4} \right) \\
 &= \frac{2\pi^2}{12} \\
 &= \frac{\pi^2}{6}.
 \end{aligned}$$

□

Exercise 2.2.3. Show directly that the functions:

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \dots$$

are mutually orthogonal in $L^2([-\pi, \pi])$.

Proof. For $n, m \neq 0$:

$$\begin{aligned}
 \int_{-\pi}^{\pi} \cos(nx) dx &= \frac{1}{n} \sin(nx) \Big|_{-\pi}^{\pi} = \frac{1}{n} \sin(n\pi) - \frac{1}{n} \sin(-n\pi) = \frac{2}{n} \sin(n\pi) = 0, \\
 \int_{-\pi}^{\pi} \sin(nx) dx &= -\frac{1}{n} \cos(nx) \Big|_{-\pi}^{\pi} = -\frac{1}{n} \cos(n\pi) + \frac{1}{n} \cos(-n\pi) = 0, \\
 \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx &= \int_{-\pi}^{\pi} \frac{\cos((n+m)x) + \cos((n-m)x)}{2} dx = 0, \\
 \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx &= \int_{-\pi}^{\pi} \frac{\cos((n-m)x) - \cos((n+m)x)}{2} dx = 0, \\
 \int_{-\pi}^{\pi} \cos(nx) \sin(mx) dx &= \int_{-\pi}^{\pi} \frac{\sin((n+m)x) + \sin((n-m)x)}{2} dx = 0.
 \end{aligned}$$

□

Exercise 2.2.4. Let $\{u_n\}_{n=1}^\infty$ be an orthonormal basis for a Hilbert space H . Let $\{v_n\}_{n=1}^\infty$ be an orthonormal seq which satisfies $\sum_{n=1}^\infty \|u_n - v_n\|^2 < 1$. Show that $\{v_n\}_{n=1}^\infty$ is also a Hilbert basis for H .

Proof. Let $x \in H$. Suppose $x \perp \{v_j\}_{j=1}^\infty$. By Parseval's Identity:

$$\begin{aligned} \|x\|^2 &= \sum_{j=1}^\infty |(x, u_j)|^2 \\ &= \sum_{j=1}^\infty |(x, u_j) - (x, v_j)|^2 \\ &= \sum_{j=1}^\infty |(x, u_j - v_j)|^2 \\ &\leq \sum_{j=1}^\infty \|x\|^2 \|u_j - v_j\|^2. \end{aligned}$$

Suppose towards contradiction $x \neq 0$. We can divide both sides of the above equation by $\|x\|^2$, giving:

$$1 \leq \sum_{j=1}^\infty \|u_j - v_j\|^2 < 1.$$

This is a contradiction, hence $x = 0$. Since we've shown $x \perp \{v_j\}_{j=1}^\infty$ implies $x = 0$, we've established $\{v_j\}_{j=1}^\infty$ as an orthonormal basis for H . $\square$

Exercise 2.2.5. Let H be a Hilbert space and $Y \neq \{0\}$ be a closed subspace, and let P denote orthogonal projection onto Y . Show that:

- (1) P is linear;
- (2) P is bounded and $\|P\| = 1$;
- (3) For all $x \in H$ we have $(Px, x) = \|Px\|^2$;
- (4) $I - P$ is orthogonal projection onto $Y^\perp$.

Proof. (a) For any $x_1, x_2 \in H$ and $\alpha \in \mathbb{C}$ we have:

$$\begin{aligned} P(x_1 + \alpha x_2) &= P(y_1 + y_1^\perp + \alpha y_2 + \alpha y_2^\perp) \\ &= P(y_1 + \alpha y_2 + (y_1 + \alpha y_2)^\perp) \\ &= y_1 + \alpha y_2 \\ &= P(x_1) + \alpha P(x_2). \end{aligned}$$

Thus T is linear.

(b) Let $x \in H$. Observe that:

$$\begin{aligned}\|Px\|^2 &= \|y\|^2 \\ &\leq \|y\|^2 + \|y^\perp\|^2 \\ &= \|y + y^\perp\|^2 \\ &= \|x\|^2.\end{aligned}$$

Squaring both sides gives $\|Px\| \leq \|x\|$. Dividing by $\|x\|$ on both sides of this inequality and taking the supremum over all $x \in H, x \neq 0$ gives $\|P\|_{\mathcal{L}(H,Y)} \leq 1$. Now let $y \in B_Y(0,1)$. Then:

$$\|Py\| = \|y\| = 1.$$

Hence $1 \in \{\|Px\| \mid x \in B_H(0,1)\}$. It must be the case that $1 \leq \|P\|_{\mathcal{L}(H,Y)}$. Therefore $\|P\|_{\mathcal{L}(H,Y)} = 1$.

(c) We will first show that P is self-adjoint and idempotent. Let $x_1, x_2 \in H$. We have:

$$\begin{aligned}(Px_1, x_2) &= (y, y_2 + y_2^\perp) \\ &= (y_1, y_2) + (y_1, y_2^\perp) \\ &= (y_1, y_2).\end{aligned}$$

$$\begin{aligned}(x_1, Px_2) &= (y_1 + y_1^\perp, y_2) \\ &= (y_1, y_2) + (y_1^\perp, y_2) \\ &= (y_1, y_2).\end{aligned}$$

Therefore $P^* = P$; i.e., P is self-adjoint. Now for any $x \in H$, notice that:

$$\begin{aligned}P(P(x)) &= P(y) \\ &= y \\ &= P(x).\end{aligned}$$

Thus $P^2 = P$. Using these two facts:

$$\begin{aligned}\|Px\|^2 &= (Px, Px) \\ &= (P^*Px, x) \\ &= (P^2x, x) \\ &= (Px, x).\end{aligned}$$

□

(d) For any $x \in H$:

$$\begin{aligned}(I - P)(x) &= Ix - Px \\ &= y + y^\perp - P(y + y^\perp) \\ &= y + y^\perp - y \\ &= y^\perp\end{aligned}$$

Exercise 2.2.6. Let H be a Hilbert Space with orthonormal basis $\{u_n\}_{n=1}^\infty$. We try to define a map:

$$T : H \rightarrow H \text{ by } Tx = \sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n$$

for a sequence $(\lambda_n) \subset \mathbb{C}$. First show that this map is a bounded linear operator if and only if (λ_n) is a bounded sequence. Then, under what condition is T self-adjoint?

Proof. Suppose $T \in \mathcal{L}(H)$. Then:

$$\begin{aligned} |\lambda_n| &= |\lambda_n| \|u_n\| \\ &= \|\lambda_n u_n\| \\ &= \|T u_n\| \\ &\leq \|T\| \|u_n\| \\ &= \|T\|. \end{aligned}$$

Taking the supremum over all $n \in \mathbb{N}$ gives $\sup_{n \in \mathbb{N}} |\lambda_n| \leq \|T\|_{\mathcal{L}(H)} < \infty$. Therefore (λ_n) is bounded. Conversely, suppose $\sup_{n \in \mathbb{N}} |\lambda_n| := M < \infty$. We must first show that T is linear. Let $x, y \in H$ and $\alpha \in \mathbb{C}$. Define $a_n := (x, u_n)$ and $b_n := (y, u_n)$. We can write:

$$\begin{aligned} x + \alpha y &= \sum_{n=1}^{\infty} a_n u_n + \alpha \sum_{n=1}^{\infty} b_n u_n \\ &= \sum_{n=1}^{\infty} (a_n + \alpha b_n) u_n. \end{aligned}$$

Applying T to $x + \alpha y$ gives:

$$\begin{aligned} T(x + \alpha y) &= \sum_{n=1}^{\infty} \lambda_n (a_n + \alpha b_n) u_n \\ &= \sum_{n=1}^{\infty} \lambda_n a_n u_n + \alpha \sum_{n=1}^{\infty} \lambda_n b_n u_n \\ &= Tx + \alpha Ty. \end{aligned}$$

Thus T is linear. Now let $x \in X$ and again define $a_n := (x, u_n)$. By Parseval's identity:

$$\begin{aligned}
 \|Tx\|^2 &= \sum_{n=1}^{\infty} |(Tx, u_n)|^2 \\
 &= \sum_{n=1}^{\infty} \left| \left(\sum_{j=1}^{\infty} \lambda_j a_n u_j, u_n \right) \right|^2 \\
 &= \sum_{n=1}^{\infty} \left| \sum_{j=1}^{\infty} (\lambda_j a_j u_j, u_n) \right|^2 \\
 &= \sum_{n=1}^{\infty} |\lambda_n a_n|^2 \\
 &\leq \sum_{n=1}^{\infty} |M|^2 |a_n|^2 \\
 &= |M|^2 \sum_{n=1}^{\infty} |a_n|^2 \\
 &= |M|^2 \|x\|^2.
 \end{aligned}$$

Rearranging and taking the square-root of both sides yields:

$$\frac{\|Tx\|}{\|x\|} \leq |M|.$$

Taking the supremum over all $x \in X$, $x \neq 0$ gives $\|T\|_{\mathcal{L}(H)} \leq |M| < \infty$. Thus $T \in \mathcal{L}(H)$.

Let $x, y \in H$. Observe that:

$$\begin{aligned}
 (Tx, y) &= \left(\sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n, y \right) \\
 &= \sum_{n=1}^{\infty} (\lambda_n (x, u_n) u_n, y) \\
 &= \sum_{n=1}^{\infty} \lambda_n (x, u_n) (u_n, y). \\
 (x, Ty) &= \left(x, \sum_{n=1}^{\infty} \lambda_n (y, u_n) u_n \right) \\
 &= \sum_{n=1}^{\infty} (x, \lambda_n (y, u_n) u_n) \\
 &= \sum_{n=1}^{\infty} \overline{\lambda_n (y, u_n)} (x, u_n) \\
 &= \sum_{n=1}^{\infty} \overline{\lambda_n} (x, u_n) (y, u_n).
 \end{aligned}$$

From this, T is self-adjoint if and only if $\lambda_n = \overline{\lambda_n}$; i.e., whenever $(\lambda_n) \subset \mathbb{R}$. $\square$

§ 2.3. Compact Operators

Definition 2.3.1. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ linear. We say T is compact if for every bounded sequence $\{x_n\}_{n=1}^{\infty} \subset X$, the sequence $\{Tx_n\}_{n=1}^{\infty}$ has a convergent subsequence. Equivalently, the operator T is said to be compact if $\overline{T(B_X(0,1))}$ is compact in Y .

Proposition 2.3.1. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ linear.

- (1) If T is compact, then T is bounded.
- (2) If $T \in \mathcal{L}(X, Y)$ and $R(T)$ is finite-dimensional, then T is compact.

Proof. (1) Let $x \in X$, $\|x\| = 1$. Then $Tx \in T(B_X(0,1)) \subseteq \overline{T(B_X(0,1))}$. Since T is compact, $\overline{T(B_X(0,1))}$ is compact in Y . Since $\overline{T(B_X(0,1))}$ is compact, it is bounded. Hence $\|Tx\| < \infty$.

(2) Since $T \in \mathcal{L}(X, Y)$, the subspace $T(B_X(0,1))$ is bounded. Since $R(T)$ is finite-dimensional, by Corollary 1.2.2 $\overline{T(B_X(0,1))}$ is closed. Hence $\overline{T(B_X(0,1))} = \overline{T(B_X(0,1))}$ is closed and bounded. By Theorem 1.2.1, $\overline{T(B_X(0,1))}$ is compact. Thus T is compact. $\square$

Example 2.3.1. Let X be a normed linear space. Then $\text{id} : X \rightarrow X$ is bounded, but not compact.

Define $T : \ell^2 \rightarrow \ell^2$ by $e^n \mapsto \frac{1}{\sqrt{n}} e^n$ and extend linearly. This map is compact, but $R(T)$ is not finite dimensional.

Theorem 2.3.1. Let X be a normed linear space and Y a Banach space. The set:

$$\{T \in \mathcal{L}(X, Y) \mid T \text{ is compact} \}$$

is closed in $\mathcal{L}(X, Y)$.